DESIGN AND EVALUATION OF GBBRPM: A GRAPH-BASED BOUNDED RISK PROPAGATION MODEL FOR COMPARATIVE PRIORITIZATION ON DIRECTED ACYCLIC GRAPHS ACROSS NETWORK DOMAINS

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ACRONYMS AND ABBREVIATIONS

DAG	Directed acyclic graph
GBBRPM	Graph-Based Bounded Risk Propagation Model
OSV	Open Source Vulnerabilities
SIR	Susceptible-infected-recovered
SIS	Susceptible-infected-susceptible
SWMM	Storm Water Management Model

ABSTRACT

[abstract (150 to 250 words).]

Keywords: [3 to 5 lowercase keywords, separated by commas]

CHAPTER I
INTRODUCTION

1.1 Background of the Study

Urban drainage networks are essential infrastructure for managing stormwater and reducing rainfall-induced flooding. Because drainage components are physically connected and flow is directional, the condition of one component can have consequences beyond the location where the disturbance is first observed. Studies on drainage malfunction, clogging, and blockage show that impaired components can reduce effective conveyance, alter flood behavior, and contribute to wider system consequences rather than remaining purely local events (Haghbin and Mahjouri, 2026), (Funke et al., 2025; Iqbal and Riaz, 2024; Wang et al., 2025). This makes drainage assessment a network problem rather than only a collection of isolated local inspections.

Previous drainage studies already demonstrate the practical value of prioritization. Complex-network and graph-based methods have been used to identify critical pipes, vulnerable components, weak links, and high-risk pathways by combining network structure with information such as hydraulic relevance, runoff, capacity, storage, discharge, failure combinations, or siltation condition (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). At the same time, computer-vision and monitoring approaches can identify local blockage, debris, or component condition at individual assets (Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023). These two research directions address different parts of the decision problem. Local monitoring can show how impaired a component is at the point of observation, while graph-based approaches can show where a component is structurally or functionally important. The literature therefore supports the distinction between local condition information and network-context prioritization rather than treating them as interchangeable outputs (Simone et al., 2022), (Dastgir et al., 2023), (Meijer et al., 2023; Zhang et al., 2025; Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023).

This distinction is important for inspection and maintenance decisions. In a network containing many connected components, decision-makers may need to determine which locations deserve earlier inspection, maintenance, or higher-fidelity analysis. Drainage studies on critical-pipe identification, weakest-link analysis, siltation chains, and multiple-pipe failure show that component importance can change when flow paths, capacity, network structure, or interacting failures are considered (Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). Consequently, two components with similar local disturbance may differ in network-level importance, while a structurally central component may not necessarily carry the highest current modeled risk under the present disturbance and operating conditions. The practical need addressed by this study is therefore not simply to detect blockage or rank structural importance, but to compare components according to a risk state that incorporates both local and propagated

network context.

The broader propagation literature shows that this modeling issue extends beyond drainage. Influence-diffusion models propagate activation, epidemic models propagate compartment or infection states, cascading-failure models propagate load or failure, attack-graph and cyber-risk approaches propagate reachable compromise or risk, and financial and infrastructure models propagate exposure, loss, stress, or related system states (Kempe et al., 2003; Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021; Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). These model families demonstrate that network propagation is well established, but they also show that the quantity being propagated differs substantially across applications. The update and aggregation rules used in each model are therefore tied to the semantics of that state rather than being automatically transferable to a bounded comparative risk-index task (Kempe et al., 2003), (Buldyrev et al., 2010), (Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018).

Across these model families, several recurring mechanisms are nevertheless evident. Topology establishes where interaction may occur; transmission or influence represents how strongly an effect transfers; susceptibility, capacity, vulnerability, or tolerance modifies how a relation or receiving context responds; current state influences later propagation; and incoming contributions require an aggregation rule consistent with the meaning of the propagated quantity (Kempe et al., 2003), (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). Drainage studies reinforce the same distinction by showing that structural measures are often augmented with hydraulic relevance, capacity, runoff, storage, discharge, or other operating information when functional importance is being assessed (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). These mechanisms provide useful building blocks, but the reviewed evidence does not clearly establish their particular integration into a lightweight directed-graph architecture whose primary output is a bounded continuous non-probabilistic comparative risk state.

The present study therefore proposes the **Graph-Based Bounded Risk Propagation Model (GBBRPM)**. GBBRPM is the name assigned in this study to one proposed model; it is not the name of an established model family. The broader research area is graph-based risk propagation and comparative network prioritization. The proposed contribution concerns the explicit integration and evaluation of existing propagation mechanisms for a bounded, non-probabilistic node-state task on directed acyclic graphs (DAGs). The term bounded describes a core property of the model rather than a drainage-specific disturbance: the model represents node risk as a normalized state constrained to the interval [0,1]. The architecture is generic in formulation, while its parameter definitions remain domain-specific. In a drainage instantiation, for example, local disturbance may represent blockage severity and susceptibility may be derived from load relative to effective capacity. A different domain may use the same core only if it can

defensibly define its own nodes, directed relations, disturbance, susceptibility, transmission, and comparative risk semantics. The study will therefore examine generic formulation through separate domain instantiations rather than presume universal empirical generalizability. The mathematical formulation will be implemented in a **GBBRPM evaluation engine**; this engine is the executable implementation, not a second model.

1.2 Statement of the Problem

The study addresses the problem of *comparative risk prioritization in directed acyclic graphs*, initially motivated by urban drainage. Existing detection and monitoring approaches can identify the condition of an individual drainage component, including visual blockage or obstruction, but these outputs are principally local to the monitored asset (Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023). A local blockage score may therefore indicate that a component is impaired without directly showing how that condition should be prioritized once upstream influence, downstream connectivity, or relation-specific operating conditions are considered.

Graph-based structural approaches provide an important second perspective because they use network topology to identify critical or vulnerable components. However, drainage studies show that structural importance and functional or operating response are not equivalent. Simone et al. and Reyes-Silva et al. show that structural centrality may diverge from hydraulically meaningful importance, while later critical-pipe and weakest-link studies augment graph structure with runoff, capacity, storage, discharge, or other domain information (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023). Siltation-risk analysis similarly combines centrality, pipeline risk, edge vulnerability, and propagation pathways rather than relying on topology alone (Zhang et al., 2025). Centrality can therefore identify structurally important locations, but it does not by itself model how a current disturbance recursively changes as it moves through relation-specific operating conditions.

Existing propagation models address network influence from other directions, but their state semantics are commonly designed for different analytical questions. Influence-diffusion models use activation states (Kempe et al., 2003), epidemic models use infection or compartment states (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021), cascading-failure models use load, capacity, or failure states (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017), attack-graph approaches use reachable compromise or probability (Sheyner et al., 2002), (Muñoz-González et al., 2017), and cyber, finance, transportation, and infrastructure models use other forms of risk, exposure, loss, or dynamic system state (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). These formulations are appropriate for their intended purposes, but they do not directly provide the bounded continuous non-probabilistic comparative risk state targeted in the present study.

Taken together, the literature reveals an architectural and integrative re-

search problem. The individual mechanisms relevant to the target task already exist: directed reachability is established in graph models; recursive current-state updating appears across contagion and cascading processes; capacity, vulnerability, susceptibility, or related response factors appear in infrastructure and risk models; and multiple forms of aggregation are used according to state semantics (Kempe et al., 2003; Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021; Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). Drainage research additionally demonstrates the practical importance of combining structural and operating information for prioritization (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). Within the reviewed evidence, however, these mechanisms are not clearly consolidated into one lightweight and interpretable architecture specifically intended to produce a bounded comparative risk state from local disturbance and propagated network context.

The central problem is how to formulate and evaluate a bounded, non-probabilistic model that combines local disturbance with recursively propagated network context on DAGs. The study will examine both the information added beyond local or structural scores and the computational and input costs of obtaining it. A change in ranking will be treated as evidence of a changed modeling basis, not automatically as improved physical prediction or operational usefulness.

The intended lightweight design will be assessed through the evaluation engine's complexity, runtime, memory, required inputs, and preprocessing requirements. Its limited state representation will be distinguished from higher-fidelity simulation, whose physical outputs answer different questions. The study will address the following research questions:

1. How can local disturbance, current upstream state, relation-specific modifiers, and bounded aggregation be formulated into GBBRPM for comparative node-risk prioritization on DAGs?

2. How can the proposed formulation be implemented reproducibly in a GBBRPM evaluation engine and presented in a prototype that preserves input validation and computational traceability?

3. How do propagation and heterogeneous relation conditions alter risk magnitudes and rankings relative to local-only scoring, uniform propagation, and a structural ranking descriptor?

4. What mathematical properties, sensitivity, ranking robustness, and reconvergence limitations characterize the proposed model?

5. What computational and input requirements characterize GBBRPM's intended lightweight design?

6. Can the same core be instantiated coherently in drainage, software-dependency, and electrical-network cases, and what applicability limits follow from each case's data and evaluation?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective is to formulate, implement, and evaluate GBBRPM as a proposed lightweight model for bounded comparative node-risk prioritization on directed acyclic graphs, and to examine its computational requirements, behavioral properties, and coherent instantiation across drainage, software-dependency, and electrical-network domains.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. formulate GBBRPM by specifying its DAG representation, bounded state semantics, local disturbance, relation-specific modifiers, recursive update rule, aggregation mechanism, assumptions, and expected mathematical properties;
2. implement a reproducible deterministic evaluation engine and an interactive research prototype that will validate inputs, execute the same versioned core, and expose traceable node states, edge contributions, and rankings;
3. evaluate the effect of propagated state and heterogeneous relation conditions on risk magnitudes and rankings relative to local-only scoring, uniform-susceptibility propagation, and a predeclared structural ranking descriptor;
4. establish analytical properties and assess computational behavior under controlled scenarios, including boundedness, monotonicity, boundary cases, sensitivity, perturbation robustness, and shared-origin reconvergence;
5. characterize the evaluation engine's computational and input requirements through complexity analysis, runtime and memory measurements, and a documented comparison of required information and modeled mechanisms; and
6. assess whether the same core formulation can be instantiated coherently in the three selected domains through documented DAG construction, parameter mappings, within-domain experiments, and explicit evidence and applicability limits.

Literature synthesis will inform the design requirements and interpretation of these objectives. The study's principal research outputs will be the model specification, evaluation engine, evaluation evidence, and research prototype.

1.4 Significance of the Study

1.4.1 Theoretical Significance

The study will contribute an explicitly specified architecture for combining local disturbance with recursively propagated network context on DAGs. Theoretical significance will lie in the declared non-probabilistic state semantics, relation-level contribution rule, bounded aggregation mechanism, and analytical account

of the model's properties and restrictions. The study will distinguish properties established by the formulation from claims that require computational or domain evidence, including the difference between bounded aggregation and correction for shared-origin dependence.

Existing propagation and risk literature supplies relevant mechanisms (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Muñoz-González et al., 2017; Petrone and Latora, 2018). The proposed contribution will be assessed as their particular integration for the specified comparative-state task, rather than treating every component as new. A common interface for domain-specific inputs will allow architectural reuse to be examined without assuming common physical semantics or universal applicability. Susceptibility and transmission will retain distinct conceptual roles, while their separate empirical identification will not be inferred from outputs that depend on their product.

1.4.2 Practical Significance

The proposed engine and prototype will provide an inspectable environment for exploring how local disturbances and relation conditions influence comparative node states. Scenario assumptions, incoming contributions, and rankings will be available for examination before further domain-specific analysis is considered. The intended practical role is transparent scenario exploration and research screening.

Drainage literature supports the use of graph-based screening alongside richer physical assessment (Dastgir et al., 2023), (Meijer et al., 2023). Potential applications of this study will include drainage screening, software-dependency analysis, and electrical-network scenario assessment, subject to defensible parameter construction and validation within each domain. Lower computational cost or a plausible ordering will not be treated as sufficient evidence of maintenance benefit, exploitation prediction, or fault prediction. Such benefits will require independent outcome evidence. The prototype will demonstrate the research workflow; operational decision-making will remain outside the verified contribution unless separately evaluated.

1.5 Scope and Limitations

1.5.1 Scope of the Study

The study will cover the formulation and implementation of GBBRPM for static comparative node-risk assessment on DAGs. It will specify bounded local disturbance, susceptibility, optional transmission, edge contributions, node-state aggregation, topological evaluation, and tie-aware ranking. Controlled networks will support investigation of mathematical behavior, mechanism effects, robustness, shared-origin reconvergence, and computational cost.

Three domain instantiations will be examined separately using documented topology and parameter mappings: drainage, software dependency, and electrical networks. The same core equations will be retained. The common object of investigation will be coherent instantiation of the architectural roles, rather than equality of physical outcomes or comparability of numerical scores

across domains. Operational drainage or event-aligned electrical evaluation will proceed only if suitable data become available.

The research prototype will support validated synthetic and user-imported DAG inputs, controlled parameter editing, execution of the versioned evaluation engine, and traceable graph and ranking outputs. Functional verification will assess numerical consistency, input handling, and presentation of the computation.

1.5.2 Limitations of the Study

The core will be restricted to DAGs and one-pass evaluation. It will not represent cycles, bidirectional feedback, physical-time evolution, or full hydraulic or electrical dynamics. A topological order is a computational dependency order, not a physical propagation timeline. The index will not be a calibrated probability, and the local aggregator will not correct shared-origin dependence. Susceptibility and transmission cannot be identified separately from output states alone.

Interpretation will depend on the provenance, coverage, and defensibility of domain inputs. Synthetic networks and simulated hydraulic outputs will not be treated as field observations. Advisory-based software mappings will not establish exploitation probabilities. Controlled disturbances applied to measured electrical loading will not establish event-response prediction. The currently available electrical baseline is not aligned with independent outcomes for the documented switching windows. Inadequate field records or event-aligned outcomes may restrict evaluation to structural instantiation and controlled behavior.

Higher-fidelity formulations may be required for feedback, shared dependencies, and physical state interactions (Buldyrev et al., 2010; Shuang et al., 2017; Petrone and Latora, 2018; Wu et al., 2026; Dastgir et al., 2024; Zhang et al., 2024). The study will report these omissions as tradeoffs rather than presume that reduced computational cost preserves predictive fidelity. Prototype checks will establish consistency in the tested configuration, not production security, multi-user readiness, usability, or operational utility.

1.6 Conceptual Framework

The conceptual framework of the study begins with local disturbance and directed acyclic network context. The current risk state at an upstream node is transformed into a relation-specific contribution according to susceptibility and, where justified, optional transmission. Multiple incoming contributions are then combined with any local disturbance at the receiving node using a bounded aggregation rule, producing an updated comparative risk state. That updated state can in turn influence later downstream relations, allowing the modeled risk state to evolve recursively through the network. The resulting node states provide the basis for comparative prioritization.

Figure 1.1 distinguishes the domain-supplied inputs, the generic bounded propagation core, and the comparative decision-support output.

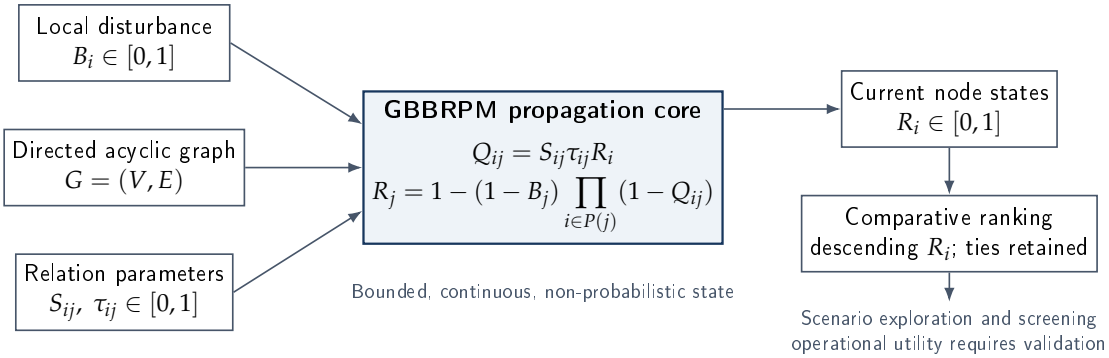


Figure 1.1. Conceptual framework of GBBRPM, distinguishing supplied domain inputs, the generic propagation core, and the comparative decision-support output.

For a directed relation $i \to j$, the generic edge contribution is:

$$Q_{ij} = S_{ij}\tau_{ij}R_i \qquad \textit{(Equation 1-1)}$$

where R_i is the current upstream normalized risk state, S_{ij} is relation-specific susceptibility, and τ_{ij} is optional transmission or influence.

The receiving-node state is:

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}) \qquad \textit{(Equation 1-2)}$$

where B_j is the local disturbance and $P(j)$ is the set of predecessors of node j. The multiplicative-complement form is used as a bounded saturating aggregation mechanism. Although it is algebraically similar to noisy-OR, it is not interpreted as a probabilistic union because the inputs are bounded risk contributions rather than calibrated probabilities.

The resulting R_j is the node's current modeled risk state. It may then serve as the upstream state for later relations, allowing risk to be recursively updated through the network. Node ranking is therefore a decision-support output derived from the resulting risk states rather than the propagation mechanism itself.

1.7 Definition of Terms

Term	Definition
Comparative risk prioritization.	Ranking components according to their modeled risk states relative to one another under the same network and parameterization.

Term	Definition
GBBRPM.	Graph-Based Bounded Risk Propagation Model, the name assigned in this study to its proposed bounded, non-probabilistic comparative-risk model on DAGs; not an established model-family name.
GBBRPM evaluation engine.	Software that will validate inputs and apply the versioned GBBRPM equations in topological order; the implementation of the model.
Directed acyclic graph (DAG).	A directed graph containing no directed cycle. This is the supported graph class for the proposed one-pass engine.
Lightweight model.	A fixed-form, single-pass DAG model without physical-time integration or training in its core. Runtime, memory, preprocessing, and input requirements will characterize this design; reduced cost does not establish physical fidelity.
Preliminary evidence.	Completed exploratory calculations or tests used to assess feasibility and limitations before protocol freezing, distinguished from independent confirmatory or outcome validation.
Normalized risk state.	A dimensionless continuous value in $[0, 1]$ representing the current modeled level of network-context risk at a node. It is not a calibrated probability of an event.
Local disturbance.	A bounded input representing risk or disturbance introduced directly at a node according to the selected domain.
Susceptibility.	A bounded relation-specific factor representing how strongly a relation or receiving context responds under the evaluated condition.
Transmission or influence.	An optional bounded factor describing how strongly influence transfers along an existing directed relation.
Edge contribution.	The relation-specific contribution generated from current upstream risk, susceptibility, and optional transmission.
Topology or reachability.	The directed network structure that determines which nodes and relations can participate in propagation.
Bounded aggregation.	The multiplicative-complement mechanism used to combine local disturbance and multiple incoming contributions while preserving the $[0, 1]$ state range.
Domain-specific instantiation.	Defining the shared node, edge, disturbance, susceptibility, and transmission roles with defensible meanings and inputs for a particular domain.
Generic formulation.	A core formulation not hard-coded to one domain. Reusability of its architecture does not establish universal empirical applicability.

CHAPTER II
REVIEW OF RELATED LITERATURE

2.1 Review Basis and Organization

This chapter synthesizes two complementary evidence streams. The first concerns urban drainage monitoring, blockage, graph-based vulnerability assessment, critical-component analysis, and hydraulic or network-level prioritization. This literature establishes the practical origin of the research problem. The second concerns foundational and modern propagation models across influence diffusion, epidemics, cascading failure, cybersecurity, finance, transportation, and infrastructure. This literature establishes the mechanisms and state semantics that can inform a general graph-risk architecture.

The review follows a two-stage synthesis. It first establishes the domain-specific problem through drainage monitoring and graph-based prioritization. It then situates that problem within the broader literature on propagation-state semantics, topology, transmission, susceptibility, recursion, aggregation, dependence, model complexity, and ranking. The final sections compare reusable architectures and domain-specific parameterization before synthesizing the research problems and the architectural gap addressed by the study.

2.2 Local Assessment and Structural Network Prioritization

The following studies establish why local condition and network importance are different analytical targets. They move from drainage connectivity to asset-level monitoring and then to structural and condition-aware ranking.

2.2.1 Urban Drainage Networks as Interconnected Systems

Urban drainage networks consist of connected conduits, junctions, inlets, screens, and other components whose behavior is influenced by flow direction, network structure, hydraulic condition, and local impairment. Blockage, clogging, siltation, structural defects, or reduced effective capacity can influence system behavior beyond the initially affected location. Numerical and network-based drainage studies show that the location and severity of impairment can alter hydraulic response, vulnerability, and the importance of connected components (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

This interconnected structure creates a practical prioritization problem. Inspection and maintenance resources are finite, so not every component can receive equal attention at the same time. Existing drainage studies explicitly frame critical-component identification as useful for targeted management, maintenance, monitoring, resilience, or dredging decisions (Simone et al., 2022), (Simone, 2023), (Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). The

practical decision problem is therefore not only to identify which components are locally impaired but also to determine which components deserve greater attention when the surrounding network is considered.

This drainage-side evidence exposes a distinction between observation and prioritization. A local blockage or defect can be observed directly, but a maintenance decision often requires a broader question: whether that component remains important after the connected network, downstream pathways, relation conditions, and converging influences are considered. This distinction between local condition and network-context priority recurs in studies that combine inspection, topology, capacity, and hydraulic relevance (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.2.2 Local Monitoring and Component-Level Condition Assessment

Computer-vision, sensor, and inspection studies have substantially improved the ability to identify local drainage condition. Iqbal et al. (2021), for example, formulate culvert blockage as an image-classification problem and evaluate several convolutional neural networks for classifying blockage from imagery. Vandaele et al. (2024) similarly develop camera-based methods for automatically detecting blockage at trash screens for maintenance and flood-management use. These approaches demonstrate that local blockage information can be generated automatically and repeatedly at monitored assets.

The outputs of these methods are valuable because they provide repeatable local condition or disturbance signals. However, a local score by itself does not determine network-context priority: it describes the monitored asset rather than the recursively accumulated state of the wider connected system (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022).

This distinction matters when two components have similar observed local conditions but occupy different positions in the network. One may lie on a relation with high operating susceptibility or feed several important downstream paths, while another may have little propagated consequence under the same network configuration. Local assessment is therefore necessary for condition awareness, but it does not by itself answer which nodes warrant greater priority after network position and operating conditions are considered (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2025).

The limitation should not be interpreted as a weakness of computer vision or sensing. Those methods solve the detection problem they were designed to solve. The unresolved issue is the interface between local condition information and network-level prioritization. This motivates the need for a propagation layer that can accept a local disturbance as input while separately representing network structure and relation-specific response (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023).

2.2.3 Graph-Based Drainage Analysis and Structural Prioritization

Drainage research has increasingly used graph theory and complex-network analysis to identify critical or vulnerable components. Simone et al. (2022) examine standard and tailored centrality metrics for vulnerability assessment, monitoring design, and contaminant or pathogenic spread. Their analysis shows

that standard degree, harmonic, betweenness, and edge-betweenness measures do not always reproduce hydraulically meaningful importance, which motivates tailored metrics that incorporate intrinsic relevance and flow direction.

Simone (2023) extends this idea using relevance-based centrality for critical-pipe identification. The method ranks nodes and pipes by combining network connectivity with intrinsic relevance rather than using plain topological centrality alone. This is significant for the present study because it demonstrates that even within a graph-only framework, useful prioritization can require information beyond raw connectivity.

Dastgir et al. (2023) propose a graph method for critical-pipe analysis based on runoff edge betweenness and capacity-related information. Their method is compared with hydrodynamic simulation and is intended to retain useful criticality information with lower computational and data requirements. The study also reports that graph-method accuracy decreases as network complexity, loop degree, and hydraulic stress increase, illustrating the tradeoff between lightweight structural approximation and richer physical behavior.

Dastgir et al. (2024) subsequently examine multiple-pipe failures in stormwater networks using graph-theoretic representations. Their work is particularly relevant to the present study because it shows that the consequences and prioritization of interacting failures cannot always be inferred from isolated single-component analysis. It therefore supports the explicit evaluation of multi-source configurations and converging effects while also reinforcing the need to distinguish graph-based screening from full hydraulic consequence prediction.

Meijer et al. (2023) develop a graph-based weakest-link method for urban drainage systems using flow-path analysis instead of repeated hydrodynamic simulation. Their work focuses on storage and discharge-capacity relationships and demonstrates that graph-based screening can make long rainfall-series analysis practically feasible. This provides further evidence that drainage research already seeks computationally efficient prioritization and vulnerability tools rather than relying exclusively on full hydraulic simulation.

Zhang et al. (2025) analyze siltation risk using complex-network concepts, centrality, pipeline risk factors, and edge vulnerability. Their study identifies critical nodes and high-risk propagation pathways, explicitly linking siltation-chain structure with maintenance and dredging prioritization. This is especially relevant because it shows that drainage risk prioritization may involve both structural importance and domain-specific condition information rather than a single metric.

Taken together, these studies establish two important points. First, prioritization is already a legitimate drainage-network task; the novelty of the present study is therefore not the act of ranking nodes or pipes. Second, the literature repeatedly augments topology with runoff, hydraulic relevance, storage/discharge capacity, edge vulnerability, siltation condition, or other operational information. Structural position and operating response are therefore related but not equivalent (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.3 Propagation Models and State Semantics

Broader network science provides several foundational propagation paradigms. Independent Cascade and Linear Threshold models represent influence through activation semantics (Kempe et al., 2003). Epidemic models such as SIR and SIS propagate compartment or infection states (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021). Cascading-failure models propagate physical load, capacity exceedance, or component failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Attack graphs represent feasible attack paths and reachable compromise states, while Bayesian attack-graph approaches introduce probabilistic compromise semantics (Sheyner et al., 2002), (Muñoz-González et al., 2017). Cyber-risk, finance, transportation, and infrastructure models introduce additional states such as accumulated risk, exposure, payment loss, systemic stress, or dynamic load (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018).

The main lesson is that the meaning of the propagated state determines the valid update and aggregation rules. Binary activation answers whether a node changes state. Probability estimates event likelihood under a probabilistic model. Physical-load models track conserved or redistributed quantities. These states are appropriate for their intended questions, but they do not automatically answer the comparative question of which node carries greater comparative network-context risk under a common parameterization (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Kavallieratos et al., 2021).

The comparison indicates that any proposed propagation model must declare whether its state represents activation, probability, physical quantity, or a non-probabilistic index. Without that declaration, similarly bounded values may be interpreted incorrectly and rankings from different model families may appear comparable when they are not (Kempe et al., 2003; Kermack and McKendrick, 1927; Muñoz-González et al., 2017; Petrone and Latora, 2018).

Table 2.1 compares the model families at the level emphasized during consultation: their structure, propagated state, update basis, analytical purpose, and limitation for a lightweight comparative-risk task. The comparison shows why no single family can be adopted merely because it uses a graph; the semantics of its state and the assumptions of its update rule must match the research question.

Table 2.1. Comparison of propagation-model families relevant to comparative network-risk prioritization.

Model family	Typical state	Propagation or update basis	Principal analytical use	Limitation for the present problem
Influence diffusion (Kempe et al., 2003)	Binary activation or threshold state	Edge influence probabilities or accumulated neighbor influence	Spread maximization and activation reach	Activation semantics do not directly represent a continuous comparative risk index.
Epidemic propagation (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021)	Susceptible, infected, and recovered compartments or infection probabilities	Contact structure, infection rate, and recovery rate	Disease spread and intervention analysis	Epidemiological compartments and rates require domain-specific assumptions not implied by network priority.
Cascading failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017)	Load, capacity, operational state, or failure state	Load redistribution, tolerance, and interdependence	Robustness and cascade consequence	Physical load conservation or failure thresholds are not generic comparative-risk semantics.
Attack graph and Bayesian attack graph (Sheyner et al., 2002; Muñoz-González et al., 2017)	Reachable compromise state or compromise probability	Feasible attack steps, conditional probabilities, or logical relations	Security-path and compromise analysis	Probabilistic outputs require calibrated conditional assumptions and dependence treatment.
Financial and infrastructure contagion (Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026)	Exposure, loss, stress, or service degradation	Dependency, exposure, vulnerability, and feedback	Systemic-risk and resilience assessment	Feedback and common exposure can require cyclic or dependence-aware dynamics.

Table 2.1. Comparison of propagation-model families relevant to comparative network-risk prioritization (continued).

Model family	Typical state	Propagation or update basis	Principal analytical use	Limitation for the present problem
Graph-based drainage screening (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025)	Centrality, criticality, capacity, or vulnerability score	Topology combined with relevance, runoff, capacity, or siltation information	Monitoring, maintenance, and critical-component screening	Scores differ in meaning and may omit recursive current-state propagation.
Hydraulic simulation and learned surrogate (Reyes-Silva et al., 2020; Dastgir et al., 2024; Zhang et al., 2024)	Flow, depth, surcharge, flood volume, or learned hydraulic state	Governing physics or simulator-trained approximation	Physical consequence analysis and prediction	Greater data, calibration, or training requirements; outputs answer a different question from lightweight comparative risk.

2.4 Mechanisms Relevant to Comparative Propagation

Having distinguished the states used by existing model families, the review examines the mechanisms that shape propagation: permitted paths, relation modifiers, recursion, aggregation, and dependence.

2.4.1 Topology, Reachability, and Transmission

Across propagation models, topology primarily determines which interactions are possible. An edge $i \rightarrow j$ indicates that influence may move from node i to node j, but the existence of the edge does not by itself define the strength of influence. Influence-diffusion models, attack graphs, cyber-risk models, and weighted infrastructure models commonly distinguish the existence of a pathway from the weight, probability, exposure, or transfer mechanism associated with that pathway (Kempe et al., 2003; Sheyner et al., 2002; Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

This distinction is particularly important for interpreting centrality. Centrality uses global graph structure and can identify structurally important nodes or edges, but it does not directly simulate the propagation of a current disturbance under current relation-specific conditions. Simone et al. (2022) provide a drainage-specific example: standard centrality metrics can rank structurally cen-

tral elements differently from hydraulically relevant elements, which motivated the incorporation of relevance and direction.

Accordingly, the literature supports treating topology as reachability while retaining transmission or influence as a conceptually separate relation-level mechanism. Where heterogeneous influence cannot be empirically justified, a neutral coefficient is more defensible than an unsupported set of edge weights (Kempe et al., 2003; Simone et al., 2022; Dastgir et al., 2023).

2.4.2 Susceptibility, Capacity, Vulnerability, and Receiver Response

A second distinction concerns transmission versus susceptibility. Transmission describes how strongly an influence transfers along a relation, whereas susceptibility describes how strongly a relation or receiving context responds under the current condition. Similar distinctions appear in models using capacity, vulnerability, tolerance, resilience, exposure, or fragility (Buldyrev et al., 2010; He et al., 2019; Kavallieratos et al., 2021; Faraji et al., 2026).

Cascading-failure models demonstrate that the same incoming load can produce different outcomes depending on component capacity or tolerance (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Cyber and infrastructure models likewise separate dependency or exposure strength from vulnerability or resilience (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023). Drainage studies that incorporate hydraulic relevance, runoff, capacity, storage/discharge relationships, siltation condition, or edge vulnerability similarly show that structural connectivity alone is not sufficient to characterize operating response (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

These patterns support keeping relation-specific susceptibility distinct from transmission rather than collapsing both concepts into one unexplained coefficient. In a drainage application, susceptibility may be linked to load relative to effective capacity, whereas heterogeneous transmission should remain neutral unless empirical evidence justifies its calibration (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.4.3 Recursive Propagation and Current-State Updating

Many propagation systems are recursive: later effects depend on the state produced by earlier propagation rather than repeatedly referencing only the original source. Epidemic models update infection state over successive interactions, cascading-failure models redistribute load after each failure, financial-contagion models update exposure or loss as conditions change, and hydraulic models update system state in response to preceding conditions (Kermack and McKendrick, 1927; Buldyrev et al., 2010; He et al., 2019; Faraji et al., 2026; Petrone and Latora, 2018).

This pattern supports current-state recursion: for a chain $A \rightarrow B \rightarrow C$, the contribution from B to C should be based on the current modeled state at B when the intended semantics permit inherited and local influence to accumulate. Conceptually, the node propagates what it currently carries rather than only the disturbance that originally entered the network (Kermack and McKendrick, 1927; Buldyrev et al., 2010; Faraji et al., 2026).

Current-state recursion is central to comparative network context. If the model repeatedly used only the original local disturbance, intermediate accumulation would be lost and downstream ranking would not reflect the evolving state of the network (Kempe et al., 2003; Buldyrev et al., 2010; Petrone and Latora, 2018).

2.4.4 Multi-Source Aggregation and Bounded State

When several upstream relations converge on the same node, their contributions require aggregation. Existing model families use different operators according to state semantics: physical loads may be summed or redistributed, threshold models accumulate influence until a threshold is crossed, and probabilities may be combined through Bayesian or noisy-OR structures when the required assumptions are satisfied (Kempe et al., 2003; Muñoz-González et al., 2017; Buldyrev et al., 2010).

For a bounded non-probabilistic comparative risk state, simple addition creates a practical difficulty because multiple contributions can exceed the declared [0,1] range. Clipping can restore the range but introduces an abrupt ceiling that loses information about how additional contributions accumulate. Multiplicative-complement aggregation is a candidate bounded, saturating operator, although its resemblance to noisy-OR algebra requires an explicit warning: it has a probabilistic interpretation only when the inputs and dependence assumptions justify one (Muñoz-González et al., 2017). The selected operator must therefore follow the declared state semantics rather than visual similarity to an established probability formula.

2.4.5 Dependence, Shared Paths, and Reconvergence

Bounded aggregation does not solve every multi-source problem. When contributions share a common origin and travel through multiple paths that later reconverge, incoming contributions may not be independent descriptions of distinct risk sources. Similar dependence problems appear in financial networks through common exposures, in infrastructure networks through shared dependencies, and in graph systems with overlapping paths (Buldyrev et al., 2010; Faraji et al., 2026; Zhou et al., 2023).

The literature therefore supports treating local aggregation and dependence as separate mechanisms. Bounding incoming contributions does not by itself correct shared-origin or common-cause dependence; reconvergence should be evaluated as an explicit limitation rather than hidden inside the aggregation rule (Buldyrev et al., 2010; Faraji et al., 2026).

This distinction is important for interpretability. A bounded output does not automatically mean that shared-origin information has been counted correctly. Dependence-aware correction remains a separate extension problem (Muñoz-González et al., 2017; Faraji et al., 2026).

2.5 Existing Ranking and Prioritization Approaches

Existing drainage and network studies rank components using different criteria. Local monitoring may rank according to observed blockage or defect severity. Centrality approaches rank according to structural importance. Weakest-link and vulnerability methods incorporate combinations of topology and domain-specific information. Failure studies rank components or combinations according to modeled consequence. Siltation studies identify high-risk nodes and propagation pathways. Hydraulic approaches may prioritize locations according to depth, surcharge, flood volume, resilience, or performance loss (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

The evidence shows that ranking is not a single modeling operation with a universal meaning. A centrality rank, blockage-severity rank, weakest-link rank, hydraulic-impact rank, and propagated-risk rank answer different questions. The semantic basis of the score must therefore be explicit before rankings can be compared (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

Table 2.2. Comparison of ranking and prioritization approaches

Approach	**Primary ranking basis**	**What the ranking answers**	**Limitation relative to the present target**
Local condition / detection (Iqbal et al., 2021; Vandaele et al., 2024; Patil et al., 2023)	Observed blockage, defect, debris, or local score	Which asset appears more locally impaired?	Does not inherently incorporate recursively propagated network context
Centrality / structural criticality (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020)	Graph position, paths, relevance, or structural interaction	Which node or edge is more structurally important?	Does not by itself represent propagation of a current disturbance under current relation conditions
Weakest-link / vulnerability methods (Dastgir et al., 2023; Meijer et al., 2023)	Structural information combined with capacity, storage, discharge, redundancy, or other domain variables	Which component most constrains or weakens system performance?	Produces specialized vulnerability or criticality semantics rather than a generic bounded recursive risk state
Siltation-chain risk (Zhang et al., 2025)	Pipeline risk, node importance, edge vulnerability, and propagation path	Which nodes and pathways are high priority for siltation management?	Domain-specific siltation semantics and parameterization

Approach	Primary ranking basis	What the ranking answers	Limitation relative to the present target
Hydraulic consequence analysis (Dastgir et al., 2024; Zhang et al., 2024)	Simulated depth, surcharge, flood volume, performance loss, or resilience	Which failures or locations cause greater physical consequence?	High-fidelity but strongly domain-specific and more data/computation intensive

The literature comparison also separates score generation from the decision-support use of a score. A method may first compute local, structural, hydraulic, or propagated values and then order components for inspection or maintenance; the resulting ranking inherits the meaning and limitations of the underlying score (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.5.1 Drainage Prioritization Evidence Matrix

The following study-level comparison makes the progression from local condition assessment to structural, hybrid, risk-path, and learned hydraulic approaches explicit.

Table 2.3. Drainage prioritization evidence matrix

Study	Primary task	Representative information used	Prioritization / output	Relevance to the present problem
Iqbal et al. (2021)	Culvert blockage classification	Image-derived visual features	Local blockage class	Provides local condition information but not propagated network context
Vandaele et al. (2024)	Trash-screen blockage monitoring	Camera imagery and learned blockage features	Asset-level blockage information for maintenance/flood management	Demonstrates operational value of automated local disturbance information
Simone et al. (2022)	Vulnerability and monitoring analysis	Topology, intrinsic relevance, flow direction	Tailored node/edge centrality ranking	Shows that raw topology alone may not represent functional importance

Study	Primary task	Representative information used	Prioritization / output	Relevance to the present problem
Simone (2023)	Critical-pipe identification	Connectivity and relevance-based centrality	Pipe criticality ranking	Supports network-aware prioritization but not current disturbance propagation
Dastgir et al. (2023)	Critical-pipe analysis	Runoff paths, edge betweenness, capacity-related information	Critical-pipe ranking compared with hydrodynamic results	Demonstrates lightweight graph screening and its accuracy/complexity tradeoff
Meijer et al. (2023)	Weakest-link analysis	Storage, discharge capacity, and graph flow paths	Subsystem weakness / flood-performance screening	Shows that prioritization may combine structure with operating-capacity information
Zhang et al. (2025)	Siltation-chain risk analysis	Pipeline risk factors, centrality, edge vulnerability, propagation paths	Critical nodes and high-risk pathways	Closest drainage precedent for propagated risk-path prioritization, but with siltation-specific semantics
Zhang et al. (2024)	Hydraulic-state surrogate prediction	Graph topology, recent hydraulic states, runoff, control/boundary information	Predicted hydraulic variables rather than a direct risk ranking	Defines the learned/high-fidelity boundary of the proposed lightweight risk-index model

The matrix shows that existing studies already provide local condition signals, graph-based criticality, capacity-aware screening, propagation pathways, and richer hydraulic state modeling. These strands remain distributed across methods with different purposes and state semantics, which creates an integrative question rather than an absence of ranking methods (Iqbal et al., 2021; Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

2.6 Computational and Input Tradeoffs

The preceding ranking approaches answer different questions. Their computational and data requirements must therefore be considered alongside their state semantics rather than treated as interchangeable measures of performance.

2.6.1 Lightweight Graph Models, Learned Surrogates, and High-Fidelity Simulation

The drainage literature contains a spectrum of model complexity. Local scores and graph indicators provide relatively inexpensive screening; graph-hydrodynamic hybrids combine structural reasoning with physical analysis; hydraulic simulators explicitly represent domain dynamics; and graph-neural-network surrogates learn simulator-derived hydraulic behavior for faster inference (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Dastgir et al. (2023) explicitly compare a graph-based critical-pipe method with hydrodynamic modeling and report substantially lower data and computational requirements for the graph approach, while also documenting accuracy loss as complexity and stress increase. Simone et al. (Simone et al., 2022) likewise position tailored complex-network metrics as complementary preliminary tools when full hydraulic simulation is more complex, data demanding, or computationally expensive.

At the learned-surrogate end of the spectrum, Zhang et al. (2024) develop a graph-neural-network surrogate for urban-drainage hydraulic prediction. The model is trained against physics-based flow-routing outputs and predicts hydraulic variables for real-time use. Physics-guided mechanisms are included to improve prediction consistency. Such a surrogate seeks to emulate hydraulic state transitions, which is a different objective from transparent graph screening or comparative index construction (Zhang et al., 2024).

The literature therefore does not imply that lightweight models are universally superior. High-fidelity models provide richer physical detail, while learned surrogates can approximate that detail after training. Lightweight graph models instead trade physical completeness for lower data requirements, simpler computation, and direct interpretability. The appropriate model depends on the decision question (Dastgir et al., 2023; Simone et al., 2022; Zhang et al., 2024).

2.6.2 Parameter Patterns in Existing Approaches

An important model-design question concerns how many parameters should be included in the generic core and its domain-specific instantiations. The literature suggests that the appropriate number cannot be fixed in advance; parameter choice follows the semantics and purpose of the model (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

Drainage approaches already illustrate this diversity. Standard centrality may use only graph structure, whereas tailored centrality can incorporate intrinsic node relevance and flow direction (Simone et al., 2022), (Simone, 2023). Dastgir et al. (2023) combine runoff-related path information with capacity-related information for critical-pipe analysis. Meijer et al. (2023) focus on subsystem storage and discharge capacities together with graph-based flow paths. Zhang

et al. (Zhang et al., 2025) combine pipeline risk factors, multiple centrality measures, edge vulnerability, and propagation-path information. Hydraulic and learned-surrogate approaches use substantially larger sets of rainfall, boundary, geometric, control, and hydraulic-state variables (Zhang et al., 2024).

These studies show that more parameters are not automatically better, just as fewer parameters are not automatically preferable. Each added variable should represent a distinct mechanism or provide empirically useful domain information. Adding several correlated or weakly justified parameters may reduce interpretability without improving the decision task (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Across the reviewed methods, a defensible reusable core retains only semantically distinct roles while allowing each domain to construct those roles from several observed variables where justified. For example, a drainage susceptibility mapping may combine hydraulic or physical attributes without requiring a generic propagation equation to absorb every raw parameter (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

Thus, the literature supports a generic core plus domain-supported parameterization rather than a universal fixed list of physical variables. The final number of drainage-specific parameters should be determined from evidence, data availability, and validation rather than from an arbitrary target count (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2024).

Table 2.4. Parameter patterns in existing approaches

Example approach	**Representative information used**	**Role in model**	**Literature-derived implication**
Tailored centrality (Simone et al., 2022; Simone, 2023)	Topology, intrinsic relevance, flow direction	Improves structural ranking	Topology can be enriched, but added variables need clear semantics
Critical-pipe graph method (Dastgir et al., 2023)	Runoff/path information and capacity-related information	Approximates criticality/flooding consequence	Shows value of separating network paths from capacity response
Weakest-link method (Meijer et al., 2023)	Storage, discharge capacity, flow paths	Screens subsystem weakness	Domain variables may be combined without redefining the generic graph concept
Siltation-chain analysis (Zhang et al., 2025)	Pipeline risk, centrality, edge vulnerability, risk path	Ranks critical nodes and paths	Multiple parameters are justified when each represents a distinct mechanism
GNN hydraulic surrogate (Zhang et al., 2024)	Hydraulic states, runoff, boundary, control variables, and graph topology	Predicts physical hydraulic states	High parameter count supports richer prediction but changes the task and data burden

2.7 Reusable Architectures and Domain-Specific Instantiation

A model can be generic in formulation without being empirically generalizable to every domain. Network science repeatedly reuses architectural ideas—nodes, directed relations, current state, transmission, response, and aggregation—while redefining their meanings for influence, epidemics, cascading infrastructure, cybersecurity, finance, transportation, and drainage (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Faraji et al., 2026; Petrone and Latora, 2018; Simone et al., 2022).

This distinction is important because the word *generalizable* can imply demonstrated performance on unseen domains. A reusable architecture supports cross-domain instantiation only when each application can defensibly map its data and semantics to the abstract roles and validate the resulting interpretation (Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

The core propagation logic can therefore remain fixed while a domain supplies its own parameter definitions and, where justified, an additional refinement layer. Drainage, cybersecurity, transportation, and supply-chain applications require different parameter definitions, evidence, and independent validation even when they share a graph-based computational pattern (Kavallieratos et al., 2021; Zhou et al., 2023; Petrone and Latora, 2018; Dastgir et al., 2023).

This structure resembles a reusable computational backbone with application-specific adaptation, but it should not be interpreted as machine-learning transfer learning. A learned surrogate transfers fitted representations or parameters; an explicit architecture instead reuses a declared computational rule whose variables must be reinterpreted and validated in each domain (Zhang et al., 2024).

2.8 Synthesis of the Research Problem and Architectural Gap

The closing synthesis connects the evidence on state semantics, mechanisms, ranking, and cost to a bounded comparative-risk design problem. It does not presume that a new model name establishes a new mathematical mechanism.

2.8.1 Synthesis of Research Problems

The reviewed evidence supports three major research problems.

First, local condition does not by itself represent network-context priority. Local monitoring can quantify condition at a component, but comparative prioritization in an interconnected system requires consideration of propagated network effects (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023).

Second, structural importance does not by itself represent current operating response. Centrality and topology use the network as context, but drainage and broader infrastructure studies repeatedly augment structure with hydraulic relevance, capacity, storage/discharge relationships, vulnerability, or other operating information (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Buldyrev et al., 2010; Faraji et al., 2026).

Third, existing propagation states are designed for different analytical

objectives. Activation, probability, infection, physical load, failure, financial loss, and hydraulic state each imply different update and aggregation rules. A bounded continuous non-probabilistic comparative risk state therefore requires an explicitly declared semantic and compatible propagation architecture (Kempe et al., 2003; Kermack and McKendrick, 1927; Sheyner et al., 2002; Buldyrev et al., 2010; Faraji et al., 2026; Zhang et al., 2024).

Together, these problems create a need for a model that can integrate local disturbance, current propagated state, and relation-specific operating response while remaining interpretable enough to explain why one node receives a higher comparative risk state than another (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

2.8.2 Architectural and Integrative Research Gap

The literature does not support claiming that graph propagation, recursive updating, capacity-based response, bounded state, edge weighting, or multi-source aggregation are individually novel. Each has precedent in established model families (Kempe et al., 2003; Buldyrev et al., 2010; Muñoz-González et al., 2017; Faraji et al., 2026; Petrone and Latora, 2018).

The gap is instead architectural and integrative. Within the reviewed drainage, infrastructure, cybersecurity, finance, transportation, cascade, epidemic, influence-diffusion, and graph-surrogate evidence, no clearly established lightweight directed-graph formulation was identified that combines all of the following under one explicit comparative-risk semantics: a bounded continuous non-probabilistic node state; recursive propagation of current upstream risk; explicit separation of directed reachability, optional transmission, and relation-specific susceptibility; local disturbance at receiving nodes; bounded aggregation of converging contributions; and explicit recognition that shared-path or common-cause dependence is distinct from local aggregation. This gap statement is a synthesis bounded to the literature set examined in this chapter (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025; Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002).

The reviewed evidence therefore establishes an architectural and integrative research gap rather than the absence of individual graph, propagation, or ranking mechanisms. A response to this gap would need to combine a clearly defined bounded state, recursive updating, relation-specific modifiers, receiving-node disturbance, convergence, and explicit dependence limits while preserving the distinction between generic computation and domain-specific evidence (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Dastgir et al., 2023; Zhang et al., 2025).

Figure 2.1 summarizes how the three reviewed evidence streams lead to this architectural and integrative gap.

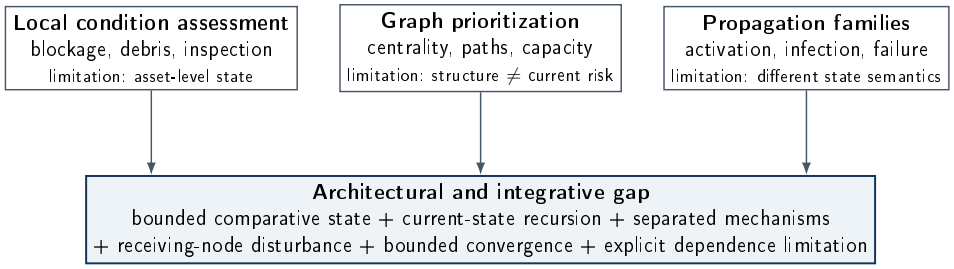


Figure 2.1. Literature-derived synthesis of existing approach families and the remaining architectural gap.

2.8.3 Chapter Synthesis

The literature establishes a clear progression from practical drainage prioritization to a broader network-risk modeling problem. Local assessment provides necessary component-level condition information, graph metrics provide structural context, and high-fidelity models provide domain-rich physical behavior. Broader propagation research contributes mechanisms for recursive updating, transmission, susceptibility, aggregation, and dependence. However, these mechanisms are distributed across models with different state semantics and purposes (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Kavallieratos et al., 2021; Petrone and Latora, 2018).

The drainage literature also demonstrates that prioritization methods already range from local condition scoring to tailored centrality, weakest-link analysis, siltation-chain risk, hydrodynamic consequence analysis, and learned hydraulic surrogates. The research opportunity is therefore not to introduce ranking itself, but to formulate a distinct comparative-risk semantics and propagation process that can explain why one node ranks above another after both local and network context are considered (Iqbal et al., 2021; Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Accordingly, the reviewed literature supports a methodological response that declares its state semantics, separates topology from relation-level response, propagates current state recursively, bounds converging contributions, and limits its claims to comparative prioritization. The specific formulation adopted by the study is presented in Chapter 3 so that this chapter remains a synthesis of prior work rather than a duplicate statement of the proposed model (Kempe et al., 2003; Buldyrev et al., 2010; Muñoz-González et al., 2017; Dastgir et al., 2023; Zhang et al., 2025).

CHAPTER III
METHODOLOGY

3.1 Research Design and Study Phases

The study will follow a quantitative model-design and computational-evaluation approach. It will formulate GBBRPM, implement the formulation in a deterministic evaluation engine, prepare documented inputs, and assess the resulting behavior and resource requirements. Three separate domain cases will examine coherent instantiation of the same equations. An interactive prototype will expose the engine's inputs, calculations, and comparative outputs.

The research will proceed through design, implementation, data preparation, evaluation, and interpretation. Literature synthesis will inform the design requirements rather than stand as a separate empirical-validation objective. The model specification and evaluation protocol will be versioned before the planned evaluation is executed. Implementation checks may reveal defects that require correction; any resulting specification or protocol change will be recorded and affected tests will be repeated.

Table 3.1. Planned objective–method–evidence alignment.

Objective	Planned method	Expected evidence or output
1: Formulate	Requirements, equations, assumptions, analytical arguments	Specified GBBRPM architecture and properties
2: Implement	Deterministic engine, schema checks, prototype integration	Versioned evaluation engine and traceable prototype
3: Compare	Paired local-only, uniform, heterogeneous, and structural conditions	Magnitude and ranking differences under controlled changes
4: Assess behavior	Boundary tests, scenario sweeps, perturbations, dependence diagnostics	Property checks, sensitivity, stability, and reconvergence limits
5: Characterize cost	Complexity analysis, separated runtime and memory measurements, input inventory	Computational and data-requirement account of lightweight design
6: Instantiate	Separate DAG construction, mappings, and within-domain analyses	Case-specific feasibility and applicability boundaries

Preliminary implementation and exploratory experiments have already been conducted. Their selected outputs are documented in Appendix D. They will inform feasibility and limitations; they will not be described as unseen confirmatory evidence. The planned study will finalize and freeze its revised specification and protocol before the next evaluation stage.

3.2 Design Requirements and Model Boundary

The formulation will address six requirements: a bounded non-probabilistic node state; simultaneous representation of local and inherited disturbance; direction-constrained influence; relation-specific modification; deterministic DAG evaluation; and traceable inputs and outputs. Existing propagation literature motivates separating the meaning of the propagated state from the mechanisms used to update it (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Muñoz-González et al., 2017; Petrone and Latora, 2018).

The study will distinguish GBBRPM, the named mathematical model, from the **GBBRPM evaluation engine**, its software implementation. The core will specify how supplied bounded inputs are combined. Domain adapters will supply their meanings and calculations; an adapter will not be permitted to silently replace the core update rule.

A domain case will be eligible only if it can provide defensible node and edge meanings and a valid DAG. A directed graph is not necessarily acyclic. Cyclic or bidirectional physical systems will not be declared compatible merely because an importer produces directed records. Any extracted subgraph, consolidation, or excluded edge will be documented together with its effect on the interpretation boundary. Modeling feedback would require a separately specified extension.

3.3 Detailed Formulation of GBBRPM

3.3.1 Graph, State, and Input Contract

The proposed graph is

$$G = (V, E), \qquad G \text{ is a directed acyclic graph.} \tag{*Equation 3-1*}$$

Nodes will represent domain entities, and an edge (i, j) will represent permitted influence from i to j. The predecessor set is $P(j) = \{i : (i, j) \in E\}$. Direction means modeled influence, which must be justified for each instantiation; it need not be physical flow in every domain.

Table 3.2. Shared notation and supplied versus derived quantities.

Symbol	Role and admissible domain	Status
$G = (V, E)$	Directed acyclic entity–relation graph	Supplied and validated
B_i	Local disturbance, $0 \leq B_i \leq 1$	Supplied or adapter-derived
S_{ij}	Relation-specific susceptibility, $0 \leq S_{ij} \leq 1$	Supplied or adapter-derived
τ_{ij}	Transmission modifier, $0 \leq \tau_{ij} \leq 1$	Supplied; neutral baseline 1
Q_{ij}	Bounded incoming contribution	Derived by the engine
R_i	Dimensionless comparative risk state, $0 \leq R_i \leq 1$	Derived by the engine
L_{ij}, C_{ij}	Compatible loading and positive capacity	Optional domain-adapter primitives

R_i will be a continuous comparative index, not a calibrated probability. Its interpretation will be conditional on the selected graph, scenario, and normalization. Values from different domains will not be placed on a common severity or likelihood scale merely because they lie in $[0, 1]$.

B_i will represent disturbance introduced directly at node i. S_{ij} will describe the response of a relation or receiving context. τ_{ij} will describe transfer along that relation. The baseline will set $\tau_{ij} = 1$ where independently defensible heterogeneous values are unavailable. A non-neutral uniform sweep will test attenuation, not empirical calibration.

3.3.2 Relation-Level Contribution and Identifiability

For a directed edge, the model will use

$$Q_{ij} = S_{ij}\tau_{ij}R_i. \qquad \text{(Equation 3-2)}$$

This specified rule makes the contribution zero when the upstream state or either modifier is zero and ensures that the contribution cannot exceed the current upstream state. It is a bounded response construction, not a physical law of transported load or energy. Branching will allow the same upstream state to influence several receivers without interpreting it as a conserved quantity that must be divided among outgoing edges.

The effective edge weight is $w_{ij} = S_{ij}\tau_{ij}$. For fixed topology and local disturbances, any admissible factor pairs with the same product give the same node outputs. Separate interpretation of susceptibility and transmission will therefore require independent measurements or justified constructions; risk outputs alone do not identify both factors.

3.3.3 Aggregation Design and Its Rationale

The complement of the receiving-node state is specified as

$$c_j = (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}), \qquad R_j = 1 - c_j. \qquad \text{(Equation 3-3)}$$

Thus the proposed update is

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}). \qquad \text{(Equation 3-4)}$$

The local complement and incoming complements jointly determine the remaining modeled margin. This construction is an algebraic choice for a bounded saturating index. Its resemblance to noisy-OR does not confer probabilistic independence or event-union semantics.

The study will explain this choice relative to alternative bounded aggregators. Sum with clipping admits cumulative contributions but reaches its cap directly; a maximum retains only the strongest input; and an arithmetic mean can fall when a weak input is added. The selected product complement permits several nonzero contributions to increase the state without exceeding one. This rationale concerns required mathematical behavior; it does not establish superior physical realism.

3.3.4 Initialization, Recursion, and Output

For a node with no predecessors, the empty product is one, giving

$$P(j) = \emptyset \quad \Longrightarrow \quad R_j = B_j. \qquad \text{(Equation 3-5)}$$

A source node may therefore contain local disturbance. Its risk is not automatically one. An isolated node follows the same rule.

Nodes will be evaluated in a topological ordering. All predecessor states will be available before a receiver is computed. Each newly computed state will immediately become the upstream state for later relations; the engine will not substitute the original B_i for an accumulated R_i. Disconnected components will be evaluated independently.

Figure 3.1 illustrates these mechanisms. Appendix A provides a complete controlled calculation. The schematic and calculation explain the model; they are not independent domain validation.

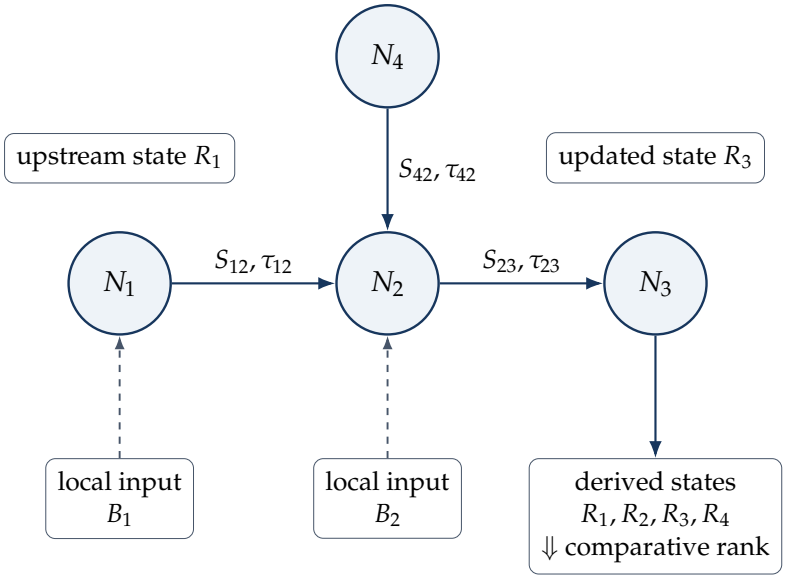


Figure 3.1. Generic directed-node example showing local inputs, relation-level modifiers, recursive state propagation, convergence at N_2, and ranking from the computed node states.

Final R_i values will be sorted in descending order. Ties will represent equal modeled standing under the declared numerical policy rather than receive an unsupported decision priority. Ranking is a derived output, not an additional propagation mechanism.

3.3.5 Analytical Properties and Dependence Boundary

Boundedness follows because every term in Equation (3-4) belongs to $[0, 1]$. For an incoming contribution $x_k = Q_{kj}$,

$$\frac{\partial R_j}{\partial x_k} = (1 - B_j) \prod_{i \neq k} (1 - x_i) \geq 0, \qquad \frac{\partial R_j}{\partial B_j} = \prod_i (1 - x_i) \geq 0. \qquad \textit{(Equation 3-6)}$$

These expressions establish local monotonicity. On a fixed DAG, global monotonicity with respect to independently increased B, S, or τ follows by induction over topological order. This argument assumes the other primitive inputs and parameter-construction rules are held fixed; it does not guarantee unchanged rankings.

If $B_j = 1$ or any incoming $Q_{ij} = 1$, the receiving state is one. A zero modifier gates its edge. By contrast, $S_{ij} = 1$ alone leaves $Q_{ij} = \tau_{ij} R_i$ and does not guarantee $R_j = 1$. The clipped L/C adapter can saturate an edge's susceptibility before the network's states saturate.

When one disturbance splits and reconverges, incoming contributions can share ancestry. Equation (3-4) will still combine them separately. Boundedness therefore does not establish dependence correction. This limitation will be evaluated explicitly rather than hidden by a stable ranking or a high receiving-node state.

3.4 Evaluation Engine and Prototype Design

3.4.1 Per-Scenario Engine Algorithm

The engine will use node records and directed adjacency lists. For each scenario it will perform the following steps:

1. Validate unique node identifiers, edge endpoints, parameter types, finite values, admissible bounds, and the declared input representation.
2. Reject self-loops, duplicate directed edges, and cyclic graphs. Generate and retain a topological order for valid input.
3. Obtain S_{ij} directly or derive it through the declared adapter. Set neutral $\tau_{ij} = 1$ only where the contract permits an omitted value.
4. For each node in topological order, compute all incoming Q_{ij}, apply Equation (3-4), and retain the node and contribution trace.
5. Rank the resulting node states with the declared tie policy and export outputs together with scenario and engine-version metadata.

The interface will not infer missing disturbances from graph position. Node B and edge S or its complete primitive pair will be required. An explicit zero disturbance will be distinguished from an unavailable observation. If both direct S and an L/C pair are provided for an edge, the final contract will reject the ambiguous representation. Capacity must be positive, load nonnegative, and units compatible.

Input violations will produce a reasoned validation failure instead of silently clipping malformed inputs. Clipping will apply only to the declared susceptibility mapping. Floating-point behavior near zero and saturation will be checked against high-precision worked cases. The engine will report the numerical tolerance used by its tests and ranking policy; any change to the arithmetic implementation will require regression comparison with the specified equations.

3.4.2 Prototype Integration and Visualization

The prototype will retain three layers: a React/Vite interface with directed graph visualization, a FastAPI validation service, and the versioned GBBRPM evaluation engine. The preliminary implementation used the `gbbrpm` v0.1.0 package; the evaluated proposal implementation will pin and record its actual version or revision.

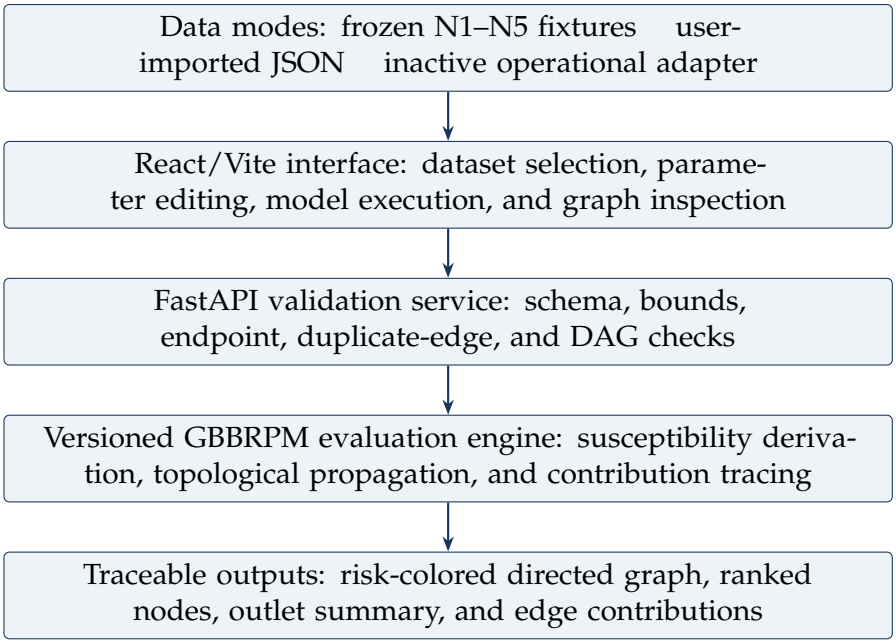


Figure 3.2. Proposed research prototype architecture separating data modes, interface, validation service, GBBRPM evaluation engine, and comparative outputs.

Synthetic and user-imported modes will pass the same validated schema to the engine. An operational adapter will remain inactive until a suitable, provenance-complete dataset is available. The interface will support parameter inspection and editing, explicit execution, graph inspection, rankings, and incoming-contribution traces.

Each evaluation response will also carry the topological order computed by the evaluation service and an ordered evaluation trace. A trace step will declare its type as source, transfer, or aggregate, together with the receiving node, that node's local disturbance B and computed risk state R, and, for a transfer step, the corresponding edge contribution Q. Trace steps will be assembled directly from quantities already computed by the engine; the service will not re-evaluate the propagation rule during trace generation, ensuring that displayed steps cannot diverge from reported results. The trace serves as a presentation record of a single evaluation rather than a redundant re-computation.

The interface will expose the analytical quantities required for that check. For a selected node, it will report relation susceptibility and incoming contributions alongside the aggregation expression of Equation (3-4); for a selected relation, it will display the numerical substitution of Equation (3-2); and for an imported network lacking physical parameters L and C, the explicit susceptibility S_{ij} will remain editable under the same unit-interval bounds as adapted edges. Selection will be coordinated between the ranking table and the graph canvas, so that choosing a reported rank position highlights the corresponding node and a rank and its topological realization are examined concurrently.

3.4.3 Evaluation Trace Presentation

Presentation of a completed evaluation will be optional and disabled by default. When enabled, the trace will be presented after execution, and a stored trace can

be replayed without issuing a further evaluation request. Because presentation operates on a returned result, it adds no computation and cannot alter numerical values.

Replay will follow a deterministic schedule compiled once from the trace. Rather than presenting the evaluation as a strictly sequential listing, the schedule will pipeline it: a node will begin receiving contributions as soon as its earliest predecessor has settled, so that independent branches advance concurrently under a single clock. Each contribution will traverse its edge only after its source has settled, and a node's displayed state will converge and be revealed only after all of its incoming contributions have arrived. The presentation will thus respect the topological evaluation order computed by the service as a dependency constraint while permitting concurrent branches to interleave in display order. Arriving contributions will be rendered as they traverse their edges, and the event boundaries of the schedule will establish discrete transition marks to which playback can seek. While a step is displayed, the interface will report the edge contribution defined by Equation (3-2) and the partially aggregated state defined by Equation (3-4), in each case converging on the value returned by the engine. The ranking derived from the evaluation will remain withheld until the replay completes or is explicitly skipped, and will be released when playback is terminated, as when a parameter is modified, the dataset is changed, or presentation is disabled. The withholding is confined to the ranking panel, so intermediate quantities are examined before final outputs are read without restricting the comparison and export views.

Replay will be interruptible and reproducible: modifying a parameter or changing the dataset will discard active playback, whereas restoring a stored replay will reproduce the identical schedule and values. Continuous motion will be suppressed, and replay will initialize in a paused state, when the environment declares a reduced-motion preference; the underlying schedule, event boundaries, and reported values remain unaffected. The reporting panel will render as an overlay layered over the visualization region so that its size cannot perturb the graph layout or trigger canvas reflows, and its vertical dimension will be adjustable by either pointer or keyboard within that constraint.

Presentation timing will remain presentational. Sequential, concurrent, or layered arrangement will not be interpreted as physical propagation speed, and no presentation option will alter the engine's topological computation or numerical outputs.

3.5 Data Sources, DAG Construction, and Parameterization

Each domain will be documented using the same template: source and version; node and edge meanings; direction and acyclicity; local disturbance; susceptibility and transmission; missingness or excluded records; and permitted output interpretation. Observed, documented, derived, assumed, synthetic, and simulated fields will be labeled separately.

3.5.1 Drainage Instantiation

Nodes and directed relations will represent the chosen drainage entities and permitted downstream influence. A load-to-capacity adapter will use

$$u_{ij} = \frac{L_{ij}}{C_{ij}}, \qquad S_{ij} = \min(1, u_{ij}), \quad L_{ij} \geq 0,\ C_{ij} > 0. \tag{*Equation 3-7*}$$

Local B_i values will represent the declared disturbance scenario or documented condition mapping. Synthetic inputs will not be called operational measurements. Operational records will be sought for topology, flow direction, specifications, loading/capacity, inspection and maintenance history, and relevant outcomes. Their adequacy will be assessed before any field-validation claim is made.

SWMM scenarios will supply an external hydraulic reference. They will be aligned to the corresponding graph, intervention, and node mapping. Acyclic downstream influence does not represent hydraulic backwater, surcharge, ponding, or redistribution. Simulated reference outputs will remain distinct from field observations.

3.5.2 Software-Dependency Instantiation

Packages will represent nodes, with edges directed from a dependency to its dependent so that modeled influence reaches the consuming package. The preliminary Express 4.18.2 case has 71 packages and 128 edges. Version-specific advisory evidence and the declared severity mapping will provide B_i; severity categories will be treated as bounded index inputs rather than calibrated exploit probabilities.

The initial case will retain $S_{ij} = 1$ and neutral baseline transmission. The uniform sensitivity sweep will use $\tau \in \{0, 0.25, 0.50, 0.75, 1.00\}$. The topology and advisory snapshots will be versioned. Cyclic dependency input will be rejected by the DAG engine; any separately proposed component-condensation approach would require its own node semantics.

Shared `qs` and `send` paths in the preliminary Express case will be identified as reconvergence rather than independent sources. The case will test mapping and parameter response, not future incident prediction.

3.5.3 Electrical Instantiation

The electrical workflow will separate full-network import verification from the observable Component 2 mechanism study using the authorized Caltech SoCal 28-Bus data (Xie et al., 2025). The preliminary full import contains 187 bus records and 177 active connections; import counts alone will not establish physical acyclicity or fault prediction.

Component 2 will be selected using observability, authorized topology, traceable meters, and DAG compatibility. Zero-impedance-connected buses will be consolidated using the documented adapter. The preliminary complete component has 13 nodes and 12 elements; exclusion of an unobserved element leaves a 12-node, 11-edge arborescence. This fit-for-purpose case will not be treated as a representative sample of distribution grids.

Synchronized phasors will provide real and reactive loading. For T aligned snapshots, downstream powers will first be aggregated within each snapshot, then converted to apparent power:

$$L_{ij} = \frac{1}{T} \sum_{t=1}^{T} \sqrt{P_{ij,t}^2 + Q_{ij,t}^2}. \qquad \textit{(Equation 3-8)}$$

The magnitude of mean P and Q will not be substituted for this mean of magnitudes. Transformer capacity will follow the documented nameplate policy; three-phase line capacity in kVA will be $3V_{\text{phase}}I_{\text{rated}}/1000$ for voltage in volts and current in amperes. Equation (3-7) will then supply electrical susceptibility.

The source and terminal-meter assignments, synchronization tolerance, unusable meters, neighboring-feeder exclusion, and unallocated residual power will be recorded. The preliminary calculation uses 24 normal-operation snapshots. Declared disturbances will be injected for controlled scenarios; they will not be relabeled as observed faults. Mains-register unit inconsistencies will be documented, with phasor-derived loading retained as the selected input.

Independent switching-event analysis will be conditional on event-aligned measurements and outcomes. Existing normal-operation coverage does not provide those outcomes. Restricted raw measurements and credentials will not be included in the public manuscript package.

3.6 Controlled Experiments and Comparative Conditions

The research will use linear, branching, converging, reconverging, and mixed DAGs to separate mechanisms before examining combined behavior. Figure 3.3 is a schematic overview; the executable fixtures, not the abbreviated diagram, will define every node and edge.

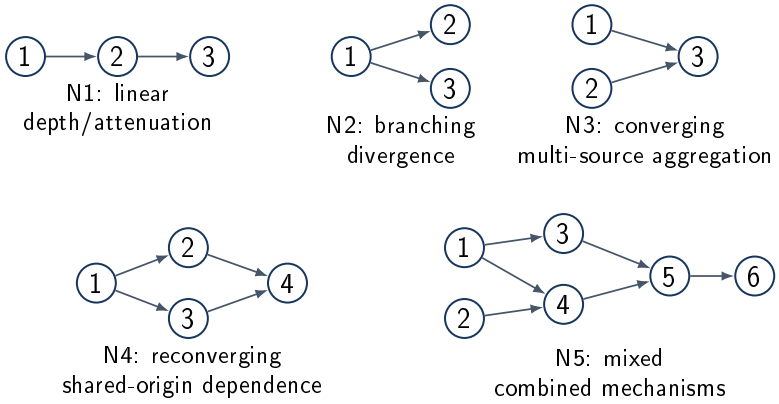


Figure 3.3. Schematic examples of the controlled DAG structures. Arrows indicate permitted influence direction; these abbreviated diagrams do not show every node in the executable N1–N5 fixtures.

The preliminary N1–N5 inventory contains 6, 8, 8, 7, and 20 nodes and 5, 7, 7, 7, and 27 edges respectively. Its recovered reconstructed inputs will be identified as such. Reproduction of historical outputs does not recover or authenticate the lost original raw dataset.

Table 3.3. Preliminary scenario configuration proposed for reviewed reruns.

Family	Configuration	Runs
Source severity	Four levels across five networks	20
Drainage L/C stress	Five load scales across five networks	25
Disturbance location	Selected historical locations	36
Multi-source configurations	Selected source subsets in N3 and N5	10
Intermediate disturbance	Three nodes and three severities across five networks	45
Total	Existing configuration; added protocols counted separately	136

The reviewed reruns will retain severity levels $B \in \{0.25, 0.50, 0.75, 1.00\}$ and load scales $\{0.50, 0.75, 1.00, 1.25, 1.50\}$ unless a documented protocol revision is needed. Location tests will set the selected source to 0.75 while clearing other local disturbances. Intermediate tests will retain baseline disturbances and add the selected local term. Exact fixtures and source subsets will be listed in the run manifest.

Paired comparison conditions will include local-only $R_j = B_j$; the same propagation rule with uniform $S = 0.25$, 0.50, or 0.75; and heterogeneous-susceptibility GBBRPM. A topology-only descriptor will use normalized directed in-degree $d_{\text{in}}(j)/(|V| - 1)$ for $|V| > 1$, with zero for a single-node graph. This descriptor represents incoming structural position, not a physical risk reference.

Changing B or S on fixed topology leaves the structural descriptor unchanged. The analysis will quantify how GBBRPM responds instead. Comparison with a topology score will establish different information dependence, not universal superiority over structural methods. Additional structural-comparison runs will be recorded separately from the preliminary 136-run count.

3.7 Properties, Sensitivity, Robustness, and Dependence

Analytical arguments will be complemented by implementation tests for zero disturbance, isolated nodes, chains, independent convergence, reconvergence, zero edge modifiers, and saturated local disturbance. Randomized admissible DAG/input cases will test the implementation against the declared properties. Passing finitely many tests will not replace the analytical argument.

Sensitivity tests will vary severity, source location, susceptibility, uniform transmission, intermediate disturbance, and source combinations, holding unrelated inputs fixed. Outputs will include node and terminal states, maximum and mean risk, risk sum, and node counts above an explicit reporting cutoff. A cutoff will be an analytical choice rather than a physical failure threshold.

The planned robustness protocol will use 600 trials at each relative perturbation magnitude $\delta \in \{0.05, 0.10, 0.20\}$ with seed 41. For this protocol, each edge susceptibility will be independently perturbed as $S'_{ij} = \min(1, \max(0, S_{ij}(1 + \epsilon_{ij})))$, where ϵ_{ij} is uniform on $[-\delta, \delta]$. Topology, disturbances, and transmission will remain fixed. This stated protocol will be versioned; earlier outputs will not be assumed identical if their generation conventions differ.

The dependence diagnostic will compare ordinary aggregation with the preliminary ancestry-aware path-pruned counterfactual. Overlapping source lineages will be grouped and their strongest incoming contribution retained; nonoverlapping groups will retain complement-product aggregation. Differences will quantify sensitivity to that particular diagnostic construction. The counterfactual will not be presented as ground truth or an independently validated dependence solution.

3.8 Computational and Input-Cost Assessment

The lightweight claim will be assessed as a resource and representation tradeoff. With adjacency-list storage and constant-cost edge operations, DAG checking, topological ordering, and propagation require

$$O(|V| + |E|), \quad \text{whereas complete comparison sorting adds } O(|V| \log |V|). \tag{Equation 3-9}$$

Graph and stored contribution-trace memory will be $O(|V| + |E|)$. The claim applies to one scenario with prepared inputs. Domain adapters, data retrieval, many repeated scenarios, and interface rendering have additional costs.

Sparse DAG sizes will initially include $|V| \in \{20, 50, 100, 250, 500, 1000\}$ with recorded edge counts, followed by a denser-series stress test at 20, 50, 100, and 250 nodes with $|E| = \lfloor 0.25|V|(|V| - 1)/2 \rfloor$. A seeded random topological ordering will restrict all edges to forward pairs, and unique pairs will be selected up to the declared edge count. Sparse cases will target three edges per node. Graphs and bounded inputs will be fixed by seed 41. Five warm-up runs and 30 measured repetitions per configuration will provide median, interquartile range, and raw timings. Graph validation/order, propagation, ranking, and total engine evaluation will be timed separately, alongside peak memory under a stated measurement boundary. Hardware, operating system, language, library versions, and timer/memory instrumentation will be recorded.

The same prepared inputs will be used for relevant local, uniform, and structural comparison implementations. Simpler baselines may remain cheaper. A required-input inventory will separately list observed and derived quantities, adapter effort, and missingness requirements. A smaller core will not imply easy data acquisition.

Higher-fidelity comparison will document the modeled quantities, omitted mechanisms, required fields, and workflow costs. A quantitative SWMM speed claim will be made only if a separately documented paired benchmark is performed with declared scenarios, simulation duration, solver settings, hardware, and timing boundaries. A hydraulic simulation and a static comparative-index evaluation remain different tasks.

3.9 Separate Domain-Case Evaluation

The drainage case will compare scenario-induced priority changes with SWMM maximum-depth and flooding changes under the aligned intervention protocol. Spearman association, priority-set overlap, and the share of positive hydraulic worsening outside the downstream region will be reported. Negative or low agreement will inform the limits of the graph index rather than be converted into evidence of validated utility.

The software case will report root and network states, root rank, and ranking agreement across the uniform transmission sweep. Advisory-derived disturbance is an input, not an independent incident label. A high root state may include shared-path influence, whose contribution is not quantified merely by stable ranking.

The observable electrical case will propose reruns of the preliminary 43-scenario configuration: one zero-disturbance reference; four root severity levels; disturbance at each of 12 nodes; five load scales; five transmission levels; 15 pairs of internal sources; and one all-internal case. Node and terminal risk, risk sums, and affected-node counts will be reported. The arborescence tests local-plus-incoming aggregation and branch coverage, not downstream reconvergence.

Cross-case synthesis will examine defensible mappings, DAG eligibility, core-rule consistency, computational requirements, and evidence limits. Domain scores, correlations, and physical outcomes will not be pooled as interchangeable performance measures. A common equation does not establish common calibration.

3.10 Prototype Functional Verification

The prototype will be checked against the versioned engine for fixture inventory, reproduction of frozen reference states within declared tolerance, valid explicit-susceptibility imports, and rejection of unknown endpoints, duplicate records, NaN and negative-infinity inputs, invalid bounds, and cycles.

Verification will treat the prototype as a presentation layer over the frozen engine, testing agreement rather than re-deriving results. The returned trace will be evaluated across all benchmark fixtures (N1–N5) for determinism and strict correspondence with reported engine outputs: it must account for each node and each edge exactly once; each step must reproduce the reported local disturbance B and state R of its node; transfer steps must reference already-evaluated sources in topological order; and each step's classification must correspond to the presence of incoming contributions. Selected receiving states will be reconstructed algebraically from displayed contributions as an independent check on trace completeness.

The replay schedule will be verified for settlement causality, which requires that no node's displayed state be revealed before every incoming contribution has landed and that each contribution begin only after its source has settled; for strictly ordered event boundaries; for valid concurrency of independent branches; and for deterministic execution that does not mutate its input. Two invariance conditions will be tested explicitly: (1) numerical results and the final

ranking must be identical across display modes and with presentation disabled, and (2) displayed quantities must agree with the engine's values at settled points, namely once a contribution has arrived and once a node has aggregated all of its incoming contributions; values between these points are presentational interpolation and are not held to this condition.

A production frontend build and the parameter-edit-and-run workflow will be checked. Presentation geometry will be inspected for containment within the visualization region under both pointer and keyboard adjustment. Interactive graph transitions, selection correspondence, and responsive layout will be verified manually in a browser because automated headless tests do not evaluate interactive canvas rendering. These checks will assess implementation consistency only; usability, accessibility, security, multi-user operation, persistence, and deployment hardening will require separate evaluation if added to scope.

3.11 Analysis, Evidence Synthesis, and Reproducibility

3.11.1 Metrics and Numerical Reporting

Paired scenario differences will be reported at node and network level. Spearman correlation will use average ranks for tied standing; a constant rank vector will be reported as undefined rather than assigned a favorable correlation. Numerical tie grouping will use a declared absolute tolerance of 10^{-12}, with each descending group's highest score as its anchor.

Top-k sets will use the nominal cutoff $k = \min(3, |V|)$ or $k = \max(1, \lceil 0.20|V| \rceil)$, depending on the declared metric. All scores within the numerical tolerance of the cutoff score will be included. The resulting set can exceed nominal k. Jaccard similarity will be $J(A, B) = |A \cap B|/|A \cup B|$. Top-three and top-20% measures will be labeled separately. These conventions will be applied consistently to planned runs and will not be retroactively attributed to every historical table.

Boundary and analytical tests will assess formal correctness; comparator differences will assess added information dependence; resource measures will assess cost; and domain analyses will support only their declared case-specific claims. Output disagreement alone will not establish better prioritization.

3.11.2 Provenance and Acquisition Contingencies

The record will include input hashes, source versions, engine revision, scenario settings, random seeds, trial counts, environment, and output files. Reconstructed historical inputs will be distinguished from their lost original raw dataset. The software topology and electrical measured loading will remain real-data inputs despite the preliminary status of their experiments.

Previously examined scenarios will remain exploratory. Any new confirmatory evaluation will specify its criteria and independent data before inspecting outcomes. If suitable agency records or event-aligned electrical evidence are unavailable at the acquisition checkpoint, the study will report that limitation and review the permitted scope with the adviser. Missing field records will not be replaced with invented operational observations.

3.12 Timeline of the Study

Figure 3.4 provides a provisional 24-week schedule spanning October 2026 to April 2027, with W1 beginning October 12, 2026. Defense, submission, and event dates will be aligned with confirmed institutional schedules. Literature gathering, initial formulation, exploratory implementation, and preliminary tests conducted before this proposal will be recorded with their actual dates rather than shown as uncompleted work.

Writing and data follow-up will overlap appropriate phases. W8 will serve as an acquisition checkpoint. Revision buffers will follow both proposal and final presentation. Dissemination will be conditional on acceptance and event scheduling rather than treated as a model-validation criterion.

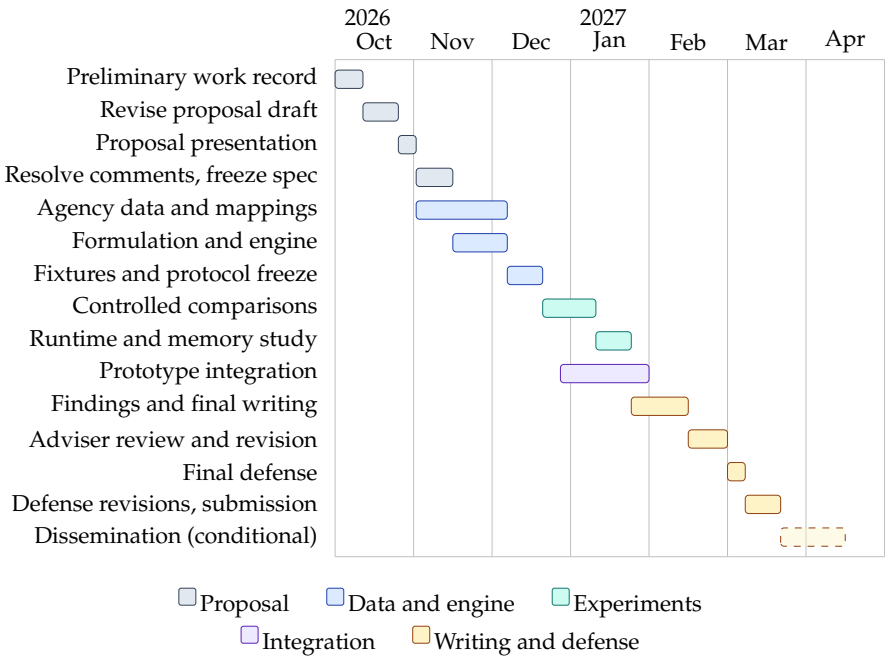


Figure 3.4. Planned schedule, October 2026 to April 2027.

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APPENDIX A
Worked Example of the Generic GBBRPM Core Architecture

A.1 Purpose and Scope of the Example

This appendix provides a transparent, step-by-step calculation of the generic Graph-Based Bounded Risk Propagation Model (GBBRPM) core architecture. The example demonstrates source-node initialization, relation-level contribution, bounded convergence, local disturbance at a receiving node, current-state recursion, and final comparative ranking. All values are controlled, dimensionless illustrative inputs. They are not drainage measurements, software observations, electrical measurements, calibrated probabilities, or evidence of empirical predictive validity.

The calculation uses the controlled directed acyclic graph shown in Figure A.1. Its directed relations are $N_1 \to N_2$, $N_4 \to N_2$, and $N_2 \to N_3$. Nodes N_1 and N_4 are sources, the two incoming paths converge at N_2, and the accumulated state at N_2 is then propagated to N_3. A valid topological evaluation order is N_1, N_4, N_2, N_3.

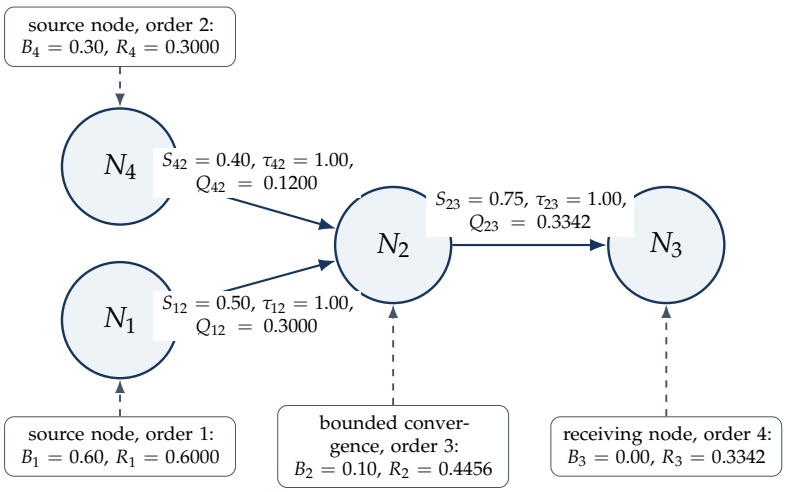


Figure A.1. Controlled four-node DAG used in the Appendix A calculation. Node callouts show evaluation order, local disturbance, and computed state; relation labels show susceptibility, neutral transmission, and incoming contribution.

A.2 Controlled Inputs

Table A.1 lists the local disturbance assigned to each node. The nonzero B_2 value is included deliberately to show that a receiving node can contain both locally introduced and inherited risk.

Table A.1. Controlled Node Inputs for the Worked Example

Node	Predecessor set $P(j)$	Local disturbance B_j
N_1	$\emptyset$	0.60
N_4	$\emptyset$	0.30
N_2	$\{N_1, N_4\}$	0.10
N_3	$\{N_2\}$	0.00

Table A.2 lists the relation-specific susceptibilities and transmission factors. Neutral transmission $\tau_{ij} = 1$ is used consistently with the frozen v1 baseline. Therefore, the worked example does not claim to evaluate heterogeneous transmission.

Table A.2. Controlled Relation Inputs for the Worked Example

Relation	Susceptibility S_{ij}	Transmission τ_{ij}	Role
$N_1 \rightarrow N_2$	0.50	1.00	First input to convergence
$N_4 \rightarrow N_2$	0.40	1.00	Second input to convergence
$N_2 \rightarrow N_3$	0.75	1.00	Recursive downstream transfer

All supplied values lie in $[0, 1]$, every relation has a valid endpoint, and the graph contains no directed cycle. The inputs therefore satisfy the declared bounded-input and DAG requirements before propagation begins.

A.3 Core Update Rules

For a directed relation $i \rightarrow j$, the incoming contribution is calculated as

$$Q_{ij} = S_{ij}\tau_{ij}R_i. \qquad \textit{(Equation A-1)}$$

where:
Q_{ij} = contribution entering node j from predecessor i;
S_{ij} = bounded relation-specific susceptibility;
τ_{ij} = bounded transmission or influence factor; and
R_i = current accumulated state of upstream node i.

The receiving-node state is calculated as

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}). \qquad \textit{(Equation A-2)}$$

where:
R_j = final bounded state of receiving node j;
B_j = local disturbance introduced directly at node j;
$P(j)$ = set of immediate predecessors of node j; and
Q_{ij} = incoming contribution from predecessor i.

The multiplicative-complement expression is a bounded saturating aggregator. It is used algebraically and is not interpreted as a probabilistic union.

A.4 Step 1: Initialize the Source Nodes

Nodes N_1 and N_4 have no predecessors. For either source node, the product over the empty predecessor set equals one. Equation A-2 therefore reduces to $R_j = B_j$:

$$
\begin{aligned}
R_1 &= 1 - (1 - B_1) \prod_{i \in \emptyset} (1 - Q_{i1}) \\
&= 1 - (1 - 0.60)(1) \\
&= 0.60, \\
R_4 &= 1 - (1 - B_4) \prod_{i \in \emptyset} (1 - Q_{i4}) \\
&= 1 - (1 - 0.30)(1) \\
&= 0.30.
\end{aligned}
\qquad \text{(Equation A-3)}
$$

Thus, a source node can contain a local disturbance. It does not require a special propagation rule: the ordinary aggregation rule produces its local state automatically through the empty-product convention.

A.5 Step 2: Calculate Contributions Entering N_2

The current source states are now available for the two relations entering N_2. Applying Equation A-1 gives

$$
\begin{aligned}
Q_{12} &= S_{12} \tau_{12} R_1 \\
&= (0.50)(1.00)(0.60) \\
&= 0.30, \\
Q_{42} &= S_{42} \tau_{42} R_4 \\
&= (0.40)(1.00)(0.30) \\
&= 0.12.
\end{aligned}
\qquad \text{(Equation A-4)}
$$

The two contributions are calculated separately because they arrive through different directed relations with different susceptibilities. Their raw sum is not used as the receiving-node state; they are combined through the bounded aggregation rule in the next step.

A.6 Step 3: Aggregate Local and Inherited Risk at N_2

Node N_2 has local disturbance $B_2 = 0.10$ and two incoming contributions. Substitution into Equation A-2 gives

$$\begin{aligned} R_2 &= 1 - (1 - B_2)(1 - Q_{12})(1 - Q_{42}) \\ &= 1 - (1 - 0.10)(1 - 0.30)(1 - 0.12) \\ &= 1 - (0.90)(0.70)(0.88) \qquad \text{(Equation A-5)} \\ &= 1 - 0.5544 \\ &= 0.4456. \end{aligned}$$

The value $R_2 = 0.4456$ represents the combined effect of the local disturbance at N_2 and the two inherited contributions. Although all three components contribute to the result, the multiplicative-complement form keeps the computed state within $[0, 1]$.

For comparison, removing the local disturbance while retaining both inherited contributions would give $1 - (1 - 0.30)(1 - 0.12) = 0.3840$. The increase from 0.3840 to 0.4456 is therefore caused by including the receiving node's own local disturbance through the same bounded rule rather than adding it after propagation.

A.7 Step 4: Propagate the Current State to N_3

The contribution from N_2 to N_3 uses the newly accumulated state $R_2 = 0.4456$, not only the original local disturbance $B_2 = 0.10$:

$$\begin{aligned} Q_{23} &= S_{23}\tau_{23}R_2 \\ &= (0.75)(1.00)(0.4456) \qquad \text{(Equation A-6)} \\ &= 0.3342. \end{aligned}$$

Because N_3 has no local disturbance and only N_2 as its predecessor, its state is

$$\begin{aligned} R_3 &= 1 - (1 - B_3)(1 - Q_{23}) \\ &= 1 - (1 - 0.00)(1 - 0.3342) \\ &= 1 - (1.00)(0.6658) \qquad \text{(Equation A-7)} \\ &= 0.3342. \end{aligned}$$

This step demonstrates current-state recursion. If the calculation incorrectly used only B_2, the relation contribution would be $(0.75)(1.00)(0.10) = 0.0750$, which would discard the risk inherited by N_2. The implemented architecture instead propagates R_2, preserving the accumulated upstream effect in the downstream evaluation.

A.8 Step 5: Produce the Comparative Ranking

After all nodes have been evaluated once in topological order, their final states are sorted in descending order. Table A.3 reports the resulting prioritization.

Table A.3. Final Node States and Comparative Ranking

Rank	Node	Final state R_i
1	N_1	0.6000
2	N_2	0.4456
3	N_3	0.3342
4	N_4	0.3000

The ranking is a comparison among modeled node states under this controlled configuration. It does not mean that N_1 has a 60% probability of failure or that the numerical spacing between ranks has a domain-independent physical meaning. If two nodes obtain exactly equal final states, the implementation retains the tie and assigns them the same risk standing rather than applying an unsupported secondary tie-breaker.

A.9 Calculation Trace and Interpretation

Table A.4 consolidates the complete evaluation trace. It also shows that every downstream calculation uses only predecessor states that have already been computed in the declared topological order.

Table A.4. Complete Topological Evaluation Trace

Order	Node	Available predecessor states	Incoming contributions	Result
1	N_1	None	Empty product	$R_1 = 0.6000$
2	N_4	None	Empty product	$R_4 = 0.3000$
3	N_2	$R_1 = 0.6000$, $R_4 = 0.3000$	$Q_{12} = 0.3000$, $Q_{42} = 0.1200$	$R_2 = 0.4456$
4	N_3	$R_2 = 0.4456$	$Q_{23} = 0.3342$	$R_3 = 0.3342$

The worked example establishes seven features of the generic architecture. First, the empty-product convention gives source nodes their local states without a separate initialization equation. Second, relation susceptibility modifies each upstream state before it enters the receiving node. Third, neutral transmission preserves the transmission interface without introducing unsupported relation differences. Fourth, multiple incoming contributions and a receiving-node disturbance are combined through one bounded rule. Fifth, the evaluator propagates the current accumulated state rather than only the original local input. Sixth, one topological pass is sufficient because the evaluated graph is acyclic. Seventh, prioritization is derived only after all node states have been calculated.

These calculations explain the mechanics of the frozen core architecture. They do not establish that the illustrative parameter values are appropriate for a particular domain. Drainage, software-dependency, and electrical-network instantiations require separate evidence for the meanings and construction of their local disturbances, susceptibilities, transmission factors, and outcome comparators.

APPENDIX B
Preliminary Worked Calculation for the Software-Dependency Case

B.1 Purpose and Selected Scenario

This appendix traces one preliminary software-domain result through the frozen GBBRPM equations. The case uses the real Express 4.18.2 dependency topology, the frozen OSV advisory snapshot, unit susceptibility on every retained dependency relation, and the controlled uniform transmission setting $\tau = 0.50$. This setting is selected because it demonstrates both intermediate-node recursion and multi-source convergence at Express. It is a sensitivity setting, not an empirically calibrated software transmission coefficient.

The original deps.dev relation points from a dependent package to the dependency it uses. For propagation, the relation is reversed so that a local disturbance in a dependency can travel toward packages that depend on it. The calculation below therefore uses dependency-to-dependent direction. Package names are shortened for readability, but every value corresponds to the exact package version retained in the preliminary software case.

B.2 Local Disturbance and Relation Inputs

The seven evidence-backed package versions have the local disturbances shown in Table B.1. Each relation uses $S_{ij} = 1$ and the selected uniform sensitivity value $\tau_{ij} = 0.50$.

Table B.1. Software Inputs Used in the Worked Calculation

Package version	Severity	B_i	Relevant outgoing relation(s)
`qs@6.11.0`	Moderate	0.50	`body-parser`, Express
`send@0.18.0`	Low	0.25	`serve-static`, Express
`body-parser@1.20.1`	High	0.75	Express
`path-to-regexp@0.1.7`	High	0.75	Express
`cookie@0.5.0`	Low	0.25	Express
`serve-static@1.15.0`	Low	0.25	Express
`express@4.18.2`	Moderate	0.50	Root dependent package

Packages without advisory-backed local disturbance have $B_i = 0$. Because their upstream states are also zero in this frozen case, their contributions to the nodes calculated below are zero and their $(1 - Q_{ij})$ terms equal one. They can therefore be omitted from the displayed products without changing the result.

B.3 Step 1: Initialize the Disturbed Dependency Nodes

The selected upstream disturbed packages have no positive incoming contribution in this scenario. Their accumulated states therefore equal their local

disturbances:

$$R_{\texttt{qs}} = 0.50, \qquad R_{\texttt{send}} = 0.25, \qquad R_{\texttt{path-to-regexp}} = 0.75, \qquad R_{\texttt{cookie}} = 0.25. \tag{\textit{Equation B-1}}$$

These are comparative normalized disturbances derived from the declared advisory-severity mapping; they are not exploitation probabilities.

B.4 Step 2: Evaluate the Intermediate Dependent Packages

The contribution from `qs` to `body-parser` is

$$\begin{aligned} Q_{\texttt{qs,body-parser}} &= S\tau R_{\texttt{qs}} \\ &= (1.00)(0.50)(0.50) \\ &= 0.25. \end{aligned} \tag{\textit{Equation B-2}}$$

Combining this incoming contribution with the local disturbance $B_{\texttt{body-parser}} = 0.75$ gives

$$\begin{aligned} R_{\texttt{body-parser}} &= 1 - (1 - 0.75)(1 - 0.25) \\ &= 1 - (0.25)(0.75) \\ &= 0.8125. \end{aligned} \tag{\textit{Equation B-3}}$$

Similarly, the contribution from `send` to `serve-static` is

$$Q_{\texttt{send,serve-static}} = (1.00)(0.50)(0.25) = 0.125. \tag{\textit{Equation B-4}}$$

The receiving package also has local disturbance $B_{\texttt{serve-static}} = 0.25$, so

$$\begin{aligned} R_{\texttt{serve-static}} &= 1 - (1 - 0.25)(1 - 0.125) \\ &= 1 - (0.75)(0.875) \\ &= 0.34375. \end{aligned} \tag{\textit{Equation B-5}}$$

Equations B-3 and B-5 demonstrate current-state recursion: the states later transmitted to Express contain both local and inherited effects.

B.5 Step 3: Calculate the Contributions Entering Express

At $\tau = 0.50$ and $S = 1$, the six nonzero contributions entering Express are shown in Table B.2.

Table B.2. Nonzero Contributions Entering Express at $\tau = 0.50$

Dependency source i	Current R_i	$Q_{i,\text{Express}} = R_i(1)(0.50)$
`body-parser`	0.812500	0.406250
`cookie`	0.250000	0.125000
`path-to-regexp`	0.750000	0.375000
`qs`	0.500000	0.250000
`send`	0.250000	0.125000
`serve-static`	0.343750	0.171875

For example, the first contribution uses the accumulated intermediate state:

$$Q_{\texttt{body-parser},\text{Express}} = (1.00)(0.50)(0.8125) = 0.40625. \qquad \textit{(Equation B-6)}$$

Using only the original $B_{\texttt{body-parser}} = 0.75$ would instead give 0.375 and would incorrectly discard the contribution already inherited from `qs`.

B.6 Step 4: Aggregate Local and Inherited State at Express

Express has its own advisory-derived local disturbance $B = 0.50$. Applying the multiplicative-complement rule to the six nonzero incoming contributions gives

$$\begin{aligned} R_{\text{Express}} &= 1 - (1 - 0.50) \times (1 - 0.40625)(1 - 0.125)(1 - 0.375) \\ &\qquad \times (1 - 0.25)(1 - 0.125)(1 - 0.171875) \\ &= 1 - (0.50)(0.59375)(0.875)(0.625) \\ &\qquad \times (0.75)(0.875)(0.828125) \\ &= 1 - (0.50)(0.1764643192) \\ &= 0.9117678404 \approx 0.9118. \end{aligned} \qquad \textit{(Equation B-7)}$$

This independently reproduces the preliminary result for Express at $\tau = 0.50$. At this setting Express becomes the highest-ranked node, while all states remain within $[0, 1]$. The example establishes traceability of the software instantiation and the effect of the controlled transmission sweep; it does not validate prediction of exploitation or operational incidents.

This calculation also exposes the dependence limitation. The `qs` state contributes both directly to Express and indirectly through `body-parser`; similarly, `send` contributes directly and through `serve-static`. The product in Equation (B-7) treats the corresponding inputs separately despite their shared origins. It reproduces the implemented v1 result, but does not correct or quantify overlap inflation.

APPENDIX C
Preliminary Worked Calculation for the Component 2 Electrical-Network Case

C.1 Purpose and Selected Scenario

This appendix traces the preliminary Component 2 root-disturbance scenario with $B_{\text{C2_N01}} = 0.75$, measured-loading scale 1.00, and neutral transmission $\tau = 1$. It first demonstrates how phasor-derived apparent power and declared capacity produce electrical susceptibility, then follows the resulting risk through the 12-node observable arborescence. The inputs are derived model-ready summaries; restricted raw phasor records and credentials are not reproduced.

All nodes other than `C2_N01` have $B_i = 0$ in this controlled scenario. Because each non-root node in the observable Component 2 graph has one predecessor, its update reduces to $R_j = Q_{ij} = S_{ij}\tau_{ij}R_i$. This simplification is specific to the selected single-source arborescence case; it does not replace the general bounded aggregation rule.

C.2 Step 1: Derive Baseline Electrical Susceptibility

For the source transformer `tr_117`, the 24 synchronized phasor snapshots produced mean downstream real power $P = 985.9568$ kW and reactive power $Q = 599.3408$ kvar. The loading is the mean of the apparent powers computed at each snapshot, rather than the apparent power computed from these two means. Writing the snapshot powers as P_t and Q_t,

$$\begin{aligned} L_{12} &= \frac{1}{24}\sum_{t=1}^{24}\sqrt{P_t^2 + Q_t^2} \\ &= 1153.8297 \text{ kVA}. \end{aligned} \qquad \text{(Equation C-1)}$$

For comparison, $\sqrt{\bar{P}^2 + \bar{Q}^2} \approx 1153.8285$ kVA. The small difference arises from the order of averaging and the nonlinear square-root operation; the snapshot-mean loading used below is unchanged.
Using the declared transformer capacity $C_{12} = 3750$ kVA, the source-edge susceptibility is

$$\begin{aligned} S_{12} &= \min\left(1, \frac{L_{12}}{C_{12}}\right) \\ &= \min\left(1, \frac{1153.8297}{3750}\right) \\ &= 0.3076879. \end{aligned} \qquad \text{(Equation C-2)}$$

For comparison, `line_408` has $L = 705.9977$ kVA and $C = 1787.94$ kVA, giving

$$S_{23} = \min\left(1, \frac{705.9977}{1787.94}\right) = 0.3948666. \qquad \textit{(Equation C-3)}$$

The same rule is applied once to every observable edge. Table C.1 lists the model-ready values needed for the complete trace.

Table C.1. Component 2 Edge Inputs Used in the Worked Calculation

Relation	Element	L (kVA)	C (kVA)	$S = L/C$
C2_N01 → C2_N02	`tr_117`	1153.8297	3750.00	0.3076879
C2_N02 → C2_N03	`line_408`	705.9977	1787.94	0.3948666
C2_N02 → C2_N11	`line_409`	375.4093	1787.94	0.2099675
C2_N03 → C2_N04	`line_415`	235.8730	1787.94	0.1319244
C2_N03 → C2_N06	`tr_121`	317.7301	1000.00	0.3177301
C2_N03 → C2_N07	`line_417`	153.8553	1787.94	0.0860517
C2_N03 → C2_N09	`line_416`	24.9112	1787.94	0.0139329
C2_N04 → C2_N05	`tr_119`	235.8730	1500.00	0.1572487
C2_N07 → C2_N08	`tr_123`	153.8553	750.00	0.2051404
C2_N09 → C2_N10	`tr_125`	24.9112	1000.00	0.0249112
C2_N11 → C2_N12	`tr_127`	375.4093	1500.00	0.2502728

All listed ratios are below one, so clipping does not alter them in the baseline scenario. The values are operating loading-to-capacity ratios, not failure probabilities.

C.3 Step 2: Initialize the Root and Evaluate the First Transfer

The root has no predecessor, so the empty-product convention gives

$$R_{\text{C2_N01}} = B_{\text{C2_N01}} = 0.75. \qquad \textit{(Equation C-4)}$$

The contribution through `tr_117` is

$$\begin{aligned} Q_{12} &= S_{12}\tau_{12}R_1 \\ &= (0.3076879)(1.00)(0.75) \\ &= 0.2307659. \end{aligned} \qquad \textit{(Equation C-5)}$$

Because $B_{\text{C2_N02}} = 0$ and the node has only this predecessor,

$$R_{\text{C2_N02}} = 1 - (1 - 0)(1 - 0.2307659) = 0.2307659. \qquad \textit{(Equation C-6)}$$

C.4 Step 3: Propagate Through the Main Branch

Using the newly accumulated state at `C2_N02`, the contribution to `C2_N03` is

$$
\begin{aligned}
Q_{23} &= (0.3948666)(1.00)(0.2307659) \\
&= 0.0911218, \\
R_{\text{C2_N03}} &= 0.0911218.
\end{aligned}
\qquad \text{(Equation C-7)}
$$

The four outgoing branches from `C2_N03` are then evaluated independently. For example, the transformer branch to `C2_N06` gives

$$
\begin{aligned}
Q_{36} &= (0.3177301)(1.00)(0.0911218) \\
&= 0.0289521, \\
R_{\text{C2_N06}} &= 0.0289521.
\end{aligned}
\qquad \text{(Equation C-8)}
$$

The parallel branch from `C2_N02` to `C2_N11` gives

$$
\begin{aligned}
R_{\text{C2_N11}} &= (0.2099675)(1.00)(0.2307659) \\
&= 0.0484533, \\
R_{\text{C2_N12}} &= (0.2502728)(1.00)(0.0484533) \\
&= 0.0121266.
\end{aligned}
\qquad \text{(Equation C-9)}
$$

These calculations show why risk decreases differently across branches: every edge applies its own measured-loading susceptibility to the current upstream state.

C.5 Step 4: Complete the Topological Trace

Table C.2 reports the complete one-pass trace. Minor differences in the displayed products can result from rounding; the evaluator uses the unrounded input values listed in the model tables.

Table C.2. Component 2 Root-Severity Trace at $B_{N01} = 0.75$

Depth	Node	Calculation using unrounded predecessor state	Final R_i
0	C2_N01	Local disturbance	0.7500000
1	C2_N02	$0.75(0.3076879)$	0.2307659
2	C2_N03	$R_{N02}(0.3948666)$	0.0911218
2	C2_N11	$R_{N02}(0.2099675)$	0.0484533
3	C2_N04	$R_{N03}(0.1319244)$	0.0120212
3	C2_N06	$R_{N03}(0.3177301)$	0.0289521
3	C2_N07	$R_{N03}(0.0860517)$	0.0078412
3	C2_N09	$R_{N03}(0.0139329)$	0.0012696
3	C2_N12	$R_{N11}(0.2502728)$	0.0121266
4	C2_N05	$R_{N04}(0.1572487)$	0.0018903
4	C2_N08	$R_{N07}(0.2051404)$	0.0016085
4	C2_N10	$R_{N09}(0.0249112)$	0.0000316

Summing the unrounded states gives

$$\sum_{i=1}^{12} R_i = 1.1860821582 \approx 1.186, \qquad \text{(Equation C-10)}$$

which reproduces the risk sum retained in the preliminary electrical case. The three nodes above the declared 0.05 affected-node threshold are `C2_N01`, `C2_N02`, and `C2_N03`; the maximum terminal-node state is $R_{\text{C2_N06}} = 0.0289521 \approx 0.029$.

The calculation establishes that measured normal-operation loading can be converted into heterogeneous bounded susceptibilities and evaluated by the unchanged GBBRPM core. The injected $B = 0.75$ is controlled rather than an observed electrical fault, so the result is mechanism and sensitivity evidence rather than event-outcome validation.

APPENDIX D
Selected Preliminary Computational Evidence

D.1 Status, Purpose, and Provenance

This appendix preserves selected completed exploratory findings that motivated the proposed evaluation. These outputs were reported in the pre-consultation full-study manuscript and have not been regenerated under every newly specified proposal convention. They demonstrate implementation feasibility and expose limitations; they are not the final results of the proposed study.

The preliminary core used recovered reconstructed validation inputs and a frozen formulation. The restored scenario inventory contained 136 runs. The five controlled fixtures contain 6, 8, 8, 7, and 20 nodes, respectively. The software topology and measured electrical loading are real-data inputs, while their disturbance mappings or scenario injections remain distinct from independent incident outcomes. A complete pre-consultation PDF and the original chapter sources are retained in the accompanying research archive.

D.2 Controlled Behavior and Comparator Differences

The baseline outlet states were 0.053760, 0.408000, 0.106814, 0.351372, and 0.708640 for N1–N5. These values are comparative indices under the declared fixtures, not failure probabilities.

Table D.1. Preliminary N5 comparator agreement with heterogeneous GBBRPM.

Comparator	Spearman ρ	Top-three Jaccard
Local-only	0.108627	0.0
Uniform $S = 0.25$	-0.042105	0.0
Uniform $S = 0.50$	0.150376	0.0
Uniform $S = 0.75$	0.696241	0.5

These differences show that the supplied relation conditions and propagated context changed the ordering. Agreement with GBBRPM is not a ground-truth accuracy measure, and the table does not establish that GBBRPM is operationally superior to the comparison scores.

D.3 Preliminary Property, Robustness, and Dependence Checks

The prior report recorded 600 randomized property-test trials with no reported failure in its boundedness, local-disturbance monotonicity, $B = 1$ boundary, and zero-susceptibility checks. This is finite implementation evidence alongside the analytical arguments, not a proof of all possible extensions or implementations.

Table D.2. Preliminary N5 robustness summary, 600 trials per perturbation level.

Perturbation	Mean Spearman	5th percentile	95th percentile	Mean top-three Jaccard	Mean top-20% Jaccard
±5%	0.979190	0.962331	0.992481	0.744500	0.776000
±10%	0.938267	0.886289	0.974436	0.546667	0.604873
±20%	0.845552	0.697067	0.935338	0.388833	0.440823

These previously reported set conventions are kept distinguishable from the explicit cutoff-tie protocol for planned runs. The top-20% metric has nominal $k = 4$ for N5. Small perturbations produced broadly stable ordering, while the highest-priority membership was more sensitive.

The ancestry-aware path-pruned diagnostic gave outlet-state differences of 0.000000 for N3, 0.099372 for N4, and 0.322637 for N5 relative to ordinary aggregation. This is sensitivity to a declared counterfactual, not a validated estimate of true physical bias. It demonstrates why a bounded output and stable ranking do not establish dependence correction.

D.4 Preliminary Runtime Evidence

The existing sparse-DAG experiment reported median timings from 1.628378 ms for 20 nodes and 60 edges to 16.650025 ms for 1000 nodes and 3000 edges, after five warm-ups and 30 repetitions. These implementation-specific timings do not provide a matched speed comparison with a hydraulic simulator, memory evidence, or a breakdown of validation, propagation, and sorting. The planned cost protocol will supply those distinctions rather than infer them from this table range.

D.5 Preliminary Hydraulic-Reference Disagreement

The prior 12-scenario SWMM comparison reported the following:

Table D.3. Preliminary comparison with simulated hydraulic worsening.

Hydraulic outcome	Mean Spearman	Top-20% Jaccard	Positive worsening outside downstream scope
Maximum depth	-0.416	0.119	71.2%
Flooding volume	-0.485	0.000	40.5%

The negative associations and low priority-set overlaps do not support hydraulic-outcome ranking equivalence under this protocol. Outside-scope worsening is consistent with mechanisms omitted by the downstream-only DAG index, but the comparison does not isolate a causal hydraulic mechanism in each

scenario. Neither disagreement nor low computation cost validates drainage inspection or maintenance utility.

D.6 Preliminary Software and Electrical Instantiation

The Express 4.18.2 case used 71 packages, 128 directed relations, and seven advisory-backed local disturbances. Root risk increased from 0.5000 at $\tau = 0$ to 0.9975 at $\tau = 1$, and its rank became first by $\tau = 0.50$. Appendix B traces the intermediate setting. Shared `qs` and `send` paths may inflate the root state; their contribution is not quantified by the stable ranking.

The observable Component 2 case used 12 nodes, 11 edges, and loading derived from 24 aligned phasor snapshots. The 43 controlled scenarios tested severity, location, load scaling, transmission, and multiple local sources. The root-disturbance case $B_{\mathrm{C2_N01}} = 0.75$ gave risk sum approximately 1.186. Appendix C provides its trace. These are injected-disturbance experiments on normal-operation measurements, not independent event-response prediction.

D.7 Preliminary Prototype Feasibility

The prior implementation report recorded a controlled-fixture inventory, frozen N5 reproduction, valid imported-graph evaluation, invalid-input rejection, and a production frontend build. Those checks support implementation feasibility within the tested configuration. The planned study will rerun and extend engine/prototype consistency checks; it will not infer usability, security, field applicability, or operational readiness from the preliminary functional tests.